Comparative Numerical and Experimental Investigation of Sloshing in Partially Filled Road Tankers of Different Geometries with Mitigation Techniques

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Declaration of Authorship

We, Sharmin Nusrat and Faria Jahan Sabrin, declare that the thesis entitled "Comparative Numerical and Experimental Investigation of Sloshing in Partially Filled Road Tankers of Different Geometries with Mitigation Techniques" and the work presented in it are our own. We, hereby confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this university.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this university or any other institution, this has been clearly stated.
- Where we have consulted the published work of others, this is always clearly attributed.
- Where we have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely our own work.
- We have acknowledged all main sources of help.
- Where the thesis is based on work done by us jointly with others, we have made clear exactly what was done by others and what we have contributed ourselves.

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Abstract

Liquid sloshing in partially filled road tankers is an important engineering concern because the movement of liquid can generate unsteady forces on the tank wall and affect vehicle stability during transportation. This study investigates the influence of tanker geometry and internal perforated baffles on sloshing behavior under controlled longitudinal excitation. Three tanker cross sections, namely rectangular, cylindrical and capsule shaped geometries, were considered to compare their sloshing response under the same filling and excitation conditions.

A numerical model was developed in ANSYS Fluent using the Volume of Fluid (VOF) method to capture the transient air water interface and obtain pressure, force and free surface behavior. The simulations were performed for both unbaffled and baffled configurations. Experimental validation was carried out using a rectangular acrylic tank mounted on a reciprocating platform, where a motor wheel connecting rod mechanism produced back and forth motion. The sloshing force was measured using a load cell connected with an HX711 amplifier and Arduino UNO.

The results indicated that tank geometry strongly affected the sloshing response. The capsule shaped tank produced the highest pressure and force values whereas the cylindrical tank showed the lowest response under the selected numerical condition. The use of perforated baffles reduced liquid motion and wall force by interrupting the direct movement of the fluid. In the rectangular tank experiment, the peak force decreased from 24 N to 15.52 N after adding baffles. The study demonstrates that appropriate tanker geometry and perforated baffle arrangement can improve sloshing control in partially filled road tanker systems.

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Nomenclature

u	Velocity vector (m/s)
ρ_a	Density of air (kg/m^3)
ρ_w	Density of water (kg/m^3)
p	Pressure (Pa)
μ	Viscosity of water (kg/ms)
g	Gravitational acceleration (m/s^2)
S_m	Imposed excitation (N)
α_w	Volume fraction of water
α_a	Volume fraction of air

Chapter 1

Introduction

Sloshing is the dynamic movement of a liquid inside a partially filled container when the container is subjected to some kind of external motion or disturbance. This disturbance may come from acceleration, deceleration, turning, vibration, tilting or any repeated motion of the container. Because the tank is not completely full, the upper surface of the liquid remains free to move. As a result, the liquid does not stay still instead, it shifts from one side to another, rises and falls and forms waves. This oscillatory movement of the liquid free surface is called sloshing. A comprehensive theoretical and practical treatment of this phenomenon was provided [1], which defined sloshing as the dynamic movement of a liquid's free surface and discussed its underlying physics, analytical formulations and engineering applications in tanks, vehicles, marine structures and aerospace systems.

1.1 Background and Motivation of Study

Road tankers play a vital role in the transportation of liquids such as water, fuel, chemicals and other industrial products. Unlike solid cargo, liquids can move freely inside a tank during transportation. When a tanker accelerates, brakes, turns or travels over uneven road surfaces, the liquid inside begins to move especially when the tank is only partially filled. This movement can significantly influence the behavior of the vehicle. The motion of the liquid creates additional forces that interact with the tanker structure and affect vehicle stability. Excessive liquid movement may lead to poor handling, reduced driver control and an increased risk of rollover during sudden maneuvers. It can also produce higher loads on the tank walls which may affect the structural integrity and long term durability of the tanker. As a result, understanding the behavior of liquid motion inside road tankers has become an important area of research for improving transportation safety and operational efficiency. The extent of liquid movement depends on several factors including the amount of liquid carried, the operating conditions of the vehicle, and the design of the tank itself. Among these factors, tank geometry is particularly important because it directly influences the flow pattern of the liquid and the

resulting forces generated inside the tank. To minimize these effects, various mitigation techniques such as internal baffles and flow control devices are commonly incorporated into tanker designs. However, the effectiveness of these techniques can vary depending on the tank shape and operating conditions.

Over the years, both numerical simulations and experimental investigations have been widely used to study liquid motion in partially filled tanks. Numerical methods provide detailed information about flow behavior and force distribution, while experimental studies help validate simulation results and capture real world fluid dynamics. Combining these approaches offers a more reliable understanding of the phenomenon and supports the development of improved tanker designs.

The motivation for this study arises from the need to better understand how different road tanker geometries influence liquid motion and how mitigation techniques can reduce its adverse effects. By conducting a comparative numerical and experimental investigation, this research seeks to evaluate the performance of different tank configurations and identify effective strategies for reducing liquid induced loads. The findings are expected to contribute to safer, more stable and more efficient road tanker operations.

Many previous studies have investigated sloshing in tanks using either numerical simulation or experimental techniques. A large number of these studies focused mainly on rectangular or cylindrical tanks, while fewer studies directly compared different road tanker shapes under the same excitation condition. Similarly, many studies examined baffles as a sloshing mitigation technique but comparative investigations involving multiple tanker geometries and different baffle arrangements are still limited.

A clear gap remains in understanding how capsule, cylindrical and rectangular tankers behave under the same back and forth motion, both without baffles and with baffles. In addition, studies that combine both numerical and experimental approaches for these tanker shapes are relatively limited. This study is intended to address that gap.

1.2 Problem Statement and Objectives of the Study

Although road tankers are widely used for transporting liquids, sloshing in partially filled tanks still creates serious challenges related to vehicle stability, internal pressure variation and

structural safety. The severity of this problem is strongly influenced by tanker geometry but rectangular, capsule shaped and cylindrical tanks are not often compared under the same numerical and experimental conditions. Moreover, while perforated baffles are commonly used to reduce liquid motion, their effectiveness depends on their orientation and interaction with the tank shape. Therefore, a clear comparison is needed to understand how different tanker geometries respond to sloshing and how longitudinal and transverse perforated baffles can reduce its effect. This study addresses this problem by numerically and experimentally investigating sloshing behavior in partially filled rectangular, capsule shaped and cylindrical road tankers with and without mitigation techniques.

The objectives of this study are :

1. To investigate the sloshing behavior of partially filled capsule shaped, cylindrical and rectangular road tankers.

2. To carry out numerical simulations for analyzing free surface motion, flow patterns and force response inside the tankers.

3. To conduct experimental investigations under controlled back and forth excitation to observe the actual sloshing behavior.

4. To compare numerical and experimental flow contours in order to evaluate the consistency of the simulated and observed sloshing patterns.

5. To measure and compare the sloshing induced force using a load cell for different tanker geometries.

6. To examine the effectiveness of longitudinal and transverse perforated baffles in reducing sloshing intensity.

7. To identify the most effective tanker geometry and baffle arrangement for improved sloshing control and safer liquid transportation.

This study covers the evaluation of liquid motion inside partially filled road tanker models under a controlled reciprocating excitation. The work is limited to three selected tanker geometries: capsule shaped, cylindrical and rectangular; with the aim of observing how the internal shape of the tanker influences the overall sloshing response.

The investigation includes both computational and laboratory based approaches. In the numerical part, the internal liquid behavior is visualized through free surface profiles, flow contours and force related responses. In the experimental part, scaled models were tested on a back and forth motion setup and the resulting sloshing force was recorded using a load cell. The comparison between these two approaches helps to assess whether the numerical model can reasonably represent the physical behavior observed during testing.

The study also considers passive control of sloshing by introducing perforated baffles inside the tanker models. Only longitudinal and transverse baffle arrangements with holes are included within the present scope. The research is therefore focused on identifying the relative influence of tanker shape and baffle orientation on liquid stability rather than covering all possible tanker designs, filling conditions, road motions or baffle geometries.

This study was carried out under controlled laboratory conditions so the findings may not fully represent all situations faced by actual road tankers during transportation. In real road conditions a tanker experiences acceleration, braking, turning, road vibration, surface irregularities and sudden disturbances at the same time. However, in this study, the tanker models were placed on a platform that moved back and forth, where rotational motion was converted into linear reciprocating motion to represent acceleration induced sloshing.

1.3 Literature Review

Sloshing in partially filled tankers is an important topic because it has a strong effect on how liquid carrying systems behave during motion. In transport applications, the movement of liquid inside the tank can create extra forces and moments which may affect the safety of the structure, the stability of the vehicle and its overall performance. The amount of sloshing depends on several factors such as the shape of the tank, the liquid fill level, the type of motion and whether any internal flow control devices are used. Over the years, many researchers have studied sloshing through experiments and numerical simulations to better understand how the liquid moves and how its effects can be reduced. Among the different topics, tank geometry and sloshing control methods have been studied the most. Therefore, the literature related to this study is discussed under two main sections and these are studies based on tank geometry and studies based on sloshing mitigation techniques.

1.3.1 Literature Review Based on Tanker Geometry

The geometry of a partially filled tanker has a strong influence on liquid sloshing because the internal shape of the container controls free surface movement, liquid mass shifting, wall impact and hydrodynamic force generation. In road transportation, this effect becomes more important as the liquid cargo is continuously disturbed by acceleration, braking, turning and road induced motion. Previous studies have shown that liquid sloshing can significantly affect the dynamic behavior of vehicles carrying liquid cargo [2]. It has also been reported that the rollover threshold of partially filled tank vehicles is strongly related to tank geometry [3]. Therefore, the selection of tanker shape is not only a structural design concern but also an important factor in vehicle stability and transportation safety.

The influence of cross sectional geometry on tanker stability has been studied for different partially filled tank configurations. Research on optimal tank geometry has shown that suitable cross sectional design can improve the static roll stability of partially filled tank vehicles [4]. Other studies have investigated the combined effect of tank cross section and road adhesion during braking in a turn conditions, where liquid movement inside the tank was found to affect vehicle response and stability [5]. Further work on lateral sloshing and roll stability indicated that simplified slosh models can be useful for predicting vehicle response, although their accuracy depends on the level of liquid motion and excitation condition [6]. A comparative study of circular, elliptical, modified oval and Reuleaux triangle cross sections also confirmed that tank shape has a clear effect on dynamic slosh loads and roll stability [7]. These findings justify the need for comparing different tanker geometries under similar operating conditions.

Rectangular tanks have been widely used in sloshing research because their simple shape makes the free surface motion easier to observe, model and validate. Numerical investigation of moving rectangular tanks has shown that liquid motion can become highly nonlinear under dynamic excitation [8]. Experimental studies on rectangular tanks have also demonstrated that sloshing pressure varies significantly at different wall locations, depending on liquid level and excitation condition [9]. In addition, combined numerical and experimental studies have shown that CFD can reasonably predict sloshing behavior in rectangular tanks when the model is properly validated [10]. Because of these advantages, rectangular tanks are often used as a reference geometry for understanding fundamental sloshing behavior.

Although rectangular tanks are useful for basic sloshing analysis, actual road tankers commonly use curved or rounded geometries because of their structural and practical benefits. Comparative studies of rectangular and cylindrical tanks subjected to harmonic and seismic excitations have shown that tank shape affects wave elevation, force response and internal liquid motion [11]. More recent work on horizontal cylindrical tanks also indicated that cylindrical geometry does not always produce lower sloshing force than rectangular geometry; rather the response depends on liquid depth, excitation frequency and natural sloshing characteristics [12]. This suggests that curved surfaces alone do not guarantee better sloshing control and each tanker geometry must be evaluated under the same operating condition.

Elliptical and semi elliptical tankers are also important in road tanker applications because their cross sectional profile can influence both roll stability and internal liquid motion. Studies on half full horizontal elliptical tanks have shown that elliptical geometry affects natural sloshing frequency and hydrodynamic response [13]. Further investigation under lateral excitation demonstrated that transient sloshing in elliptical tankers can generate complex free surface motion and force variation [14]. These studies provide useful theoretical understanding of elliptical tank behavior but most of them focus on idealized or analytical models rather than direct comparison with rectangular and capsule shaped tankers under the same experimental condition.

The role of tanker geometry becomes more significant when sloshing mitigation devices are introduced. Perforated baffles have been investigated as a passive method for reducing liquid motion by interrupting the flow and dissipating wave energy [15]. Numerical studies on T-shaped baffles in rectangular tanks have shown that baffle height and liquid level strongly influence free surface deformation and pressure response [16]. Other studies comparing perforated and imperforate baffles have reported that perforated baffles can reduce sloshing pressure while allowing controlled liquid passage through the baffle openings [17]. These findings indicate that the performance of a baffle depends not only on its own configuration but also on the geometry of the tank in which it is placed.

Recent numerical and experimental studies on tank vehicles with baffle configurations have further shown that validated CFD models can be useful for evaluating anti sloshing performance in transport tankers [18]. However, many previous studies have focused either on a single tank shape or on baffle performance alone. A direct comparison among rectangular, capsule shaped and cylindrical tanker models under the same numerical and experimental

conditions is still limited. Moreover, the combined effect of tanker geometry and perforated longitudinal or transverse baffle arrangements has not been sufficiently addressed. Therefore, the present study focuses on comparing these selected tanker geometries and evaluating the effectiveness of perforated baffles for reducing sloshing response.

1.3.2 Literature Review Based on Sloshing Mitigation Techniques

Since liquid sloshing can generate large dynamic forces, severe free surface oscillation and sudden impact pressure in partially filled tanks, different mitigation techniques have been investigated to reduce its effect. Among the available methods, baffles are considered one of the most practical solutions because they can be installed inside the tank without requiring complex external control systems. The main function of a baffle is to interrupt liquid motion, reduce wave propagation and dissipate the kinetic energy of the moving fluid. As a result, the amplitude of sloshing and the force acting on the tank wall can be reduced.

Early and recent studies have shown that the effectiveness of baffles depends strongly on their height, position, orientation and interaction with the liquid free surface. Numerical investigation of a two dimensional rectangular tank with a vertical baffle showed that increasing the baffle height can reduce free surface elevation and pressure variation, although the flow separation and vortex generation near the baffle tip also influence the damping mechanism [19]. A later three dimensional study also confirmed that baffle height is an important design parameter, where a critical height can prevent the liquid from reaching the tank roof and reduce roof impact under lateral excitation [20]. These findings indicate that baffle design cannot be selected randomly; its dimension must be related to the liquid level and excitation condition.

Different baffle arrangements have also been studied for curved and cylindrical tanks. Experimental investigation of ring baffles in a rigid cylindrical tank showed that ring type arrangements can effectively reduce sloshing loads over a range of rolling frequencies and filling depths [21]. This is important because cylindrical and capsule shaped tanks are commonly used in liquid transportation and their curved geometry may require different internal mitigation arrangements compared with rectangular tanks. The results suggest that the same baffle concept may not perform equally in all tank shapes which supports the need to evaluate baffle performance together with tanker geometry.

Perforated, porous and permeable baffles have received considerable attention because they provide flow resistance while still allowing controlled liquid passage. Experimental work on vertical baffles of different configurations showed that baffle shape and arrangement can significantly affect pressure suppression under different excitation frequencies [22]. Studies on perforated screens also showed that the opening ratio and hydrodynamic resistance of the screen influence both wave damping and force reduction [23]. Unlike a fully solid baffle, a perforated or porous baffle can reduce excessive pressure build up by allowing part of the liquid to pass through the holes. This makes perforated baffles especially relevant for transport tankers where both sloshing reduction and practical structural design are important.

More recent numerical studies have focused on special baffle shapes and porous arrangements. A slot baffle design was found to damp sloshing by using the energy of the moving liquid itself, producing additional energy dissipation near resonance conditions [24]. A horizontal porous baffle was also shown to reduce sloshing energy and wall pressure in a swaying rectangular tank [25]. Similarly, numerical analysis of impermeable and permeable vertical baffles demonstrated that permeability has a strong influence on sloshing suppression and that an appropriate permeability value can improve damping performance [26]. These studies show that the presence of holes or permeability in a baffle may improve sloshing control if the opening size and flow resistance are properly selected.

Porous media have also been used as an alternative mitigation technique. Analytical, numerical and experimental investigation of a rectangular tank with porous media showed that porous material can significantly change the dynamic characteristics of sloshing, especially near resonance [27]. However, porous media may be more difficult to implement inside transport tankers because they can increase maintenance requirements, affect cleaning and reduce usable tank volume. Therefore, perforated baffles offer a more practical compromise between damping performance and operational feasibility.

Floating and movable baffles have been proposed as advanced mitigation techniques. Numerical study of a vertical floating rigid baffle showed that floating baffles can reduce sloshing over a wider range of filling conditions compared with some fixed baffle arrangements [28]. Another study on moving baffles under microgravity conditions showed that spring supported baffles can reduce decay time and fluid kinetic energy more effectively than fixed baffles under certain excitation conditions [29]. Although these methods are technically promising, they involve additional moving components, which may make them less suitable

for simple road tanker applications where durability, fabrication and maintenance are major concerns.

Other innovative approaches have also been developed by combining sloshing mitigation with energy harvesting. A liquid tank with built in buoys was proposed to absorb energy from sloshing waves through a power take off system [30]. Another study used a deformable piezoelectric energy harvester mounted inside a liquid tank, where the device both harvested wave energy and reduced peak hydrodynamic loading like a baffle [31]. These approaches show that sloshing energy can be utilized rather than only suppressed. However, such systems are more complex than ordinary baffles and are not yet as practical for conventional road tanker design.

From the reviewed studies, it is clear that baffles remain the most widely applicable and practical method for sloshing mitigation. Solid, perforated, porous, ring, floating and movable baffles have all shown potential in reducing free surface motion, pressure fluctuation and sloshing induced forces. However, most studies have focused on a single tank shape, a specific baffle type or idealized excitation conditions. Limited work has directly compared the effectiveness of longitudinal and transverse perforated baffles in different road tanker geometries under the same numerical and experimental conditions. Therefore, the present study addresses this gap by evaluating perforated longitudinal and transverse baffle arrangements in rectangular, capsule shaped and cylindrical tanker models.

Table 1.1: Prior studies and present study on sloshing effects in fluid carrying tankers, mitigation techniques and findings

Authors	Tanker shape	Methodology	Mitigation Techniques	Key Findings
Kolaei et al. [7]	Elliptical, modified oval and Reuleaux triangle cross sections	Numerical	None	Tank cross section strongly affected slosh load and roll stability.
Akyildiz and Ünal [10]	Three dimensional rectangular tank	Numerical and experimental	None	CFD results agreed well with experimental pressure and free surface behavior.
Chen et al. [11]	Rectangular and cylindrical tanks	Numerical	None	Tank shape changed wave elevation and force response under excitation.
Xue et al. [15]	Rectangular tank	Experimental and numerical	Perforated baffle	Perforated baffles reduced sloshing pressure and free surface fluctuation.
Ünal et al. [16]	Two dimensional rectangular tank	Numerical	T shaped baffle	Baffle height above 80% of liquid level showed strong sloshing suppression.

Akyildiz [19]	Two dimensional rectangular tank	Numerical	Vertical baffle	Higher vertical baffle reduced wave elevation and pressure variation.
Akyildiz et al. [21]	Cylindrical tank	Experimental	Ring baffles	Ring baffles reduced sloshing loads in cylindrical tanks.
Liu et al. [32]	Vertical cylindrical tank	Experimental	None	Sloshing became complex near the first and second natural modes.
Ayiehfor [33]	Cylindrical and elliptical storage tanks	Numerical and experimental	Baffles with different configurations	Distributed hole baffles improved damping during emergency braking.
Chang et al. [34]	Rectangular tank with two immiscible liquids	Experimental	Vertical baffles	Vertical baffles reduced first mode internal surface sloshing.
Present study	Rectangular, cylindrical and capsule shaped tankers	Numerical and experimental	Perforated longitudinal and transverse baffles	Compares geometry and baffle effects under the same excitation condition.

1.4 Thesis Organization

This thesis is organized into six chapters. **Chapter 1** introduces the research topic by presenting the background and motivation of the study, the problem statement, the objectives, the scope and limitations and the relevant literature related to sloshing in partially filled road tankers. This chapter also identifies the research gap and explains the need for comparing different tanker geometries and baffle arrangements.

Chapter 2 describes the methodology and research design followed in this study. It includes the overall research approach, details of the experimental setup, model specifications, test conditions and measurement procedure. The numerical simulation setup is also presented in this chapter, including the governing approach, boundary conditions, mesh related considerations, solver settings and validation process. In addition, the use of engineering tools, data collection methods and data processing techniques are discussed.

Chapter 3 presents the results and discussion of the study. The numerical and experimental results are arranged using suitable figures, contours and comparative plots. The sloshing behavior of rectangular, capsule shaped and cylindrical tankers is analyzed in terms of free surface motion, flow pattern and sloshing force. The effectiveness of longitudinal and transverse perforated baffles is also discussed by comparing the sloshing response with and without mitigation arrangements.

Chapter 4 discusses the broader impacts of the research. This chapter explains how the study is related to health, safety, social, legal and environmental considerations. Since sloshing in road tankers can affect vehicle stability and transportation safety, the possible safety relevance of the work is highlighted. The chapter also addresses ethical responsibilities, proper use of references and the importance of maintaining academic integrity throughout the research.

Chapter 5 explains how the present study addresses a complex engineering problem. The chapter discusses the engineering challenges associated with sloshing such as fluid motion, tanker geometry, experimental measurement, numerical modelling, validation and practical design constraints. It also relates the work to broader engineering problem solving requirements by showing how multiple technical factors are considered together in the investigation.

Chapter 6 concludes the thesis by summarizing the major findings of the numerical and experimental study. The conclusions are presented in relation to the stated objectives, including the comparison of tanker geometries and the performance of perforated baffles. This chapter also presents the limitations of the study and provides recommendations for future work such as considering different filling levels, road conditions, baffle designs and full scale tanker models.

Chapter 2

Methodology and Research Design

A structured methodology was adopted to study the sloshing behavior of partially filled road tanker models with different geometries. The work combined experimental testing and numerical simulation so that the actual liquid motion and measured force response could be compared with the simulated results. The main steps included selecting the tanker shapes, preparing scaled models, developing the reciprocating motion setup, carrying out simulations and experiments, collecting data and comparing the outcomes. This method allowed the influence of tanker geometry and perforated baffle arrangement on sloshing reduction to be evaluated in a clear and organized way.

2.1 Research Approach

The present study follows a combined numerical and experimental research approach. Numerical simulation was used to observe the free surface motion, internal flow pattern and force response inside the tanker models while experimental testing was carried out to measure the actual sloshing force under controlled back and forth excitation. This combined approach was selected because sloshing is a complex free surface flow problem and simulation results need experimental comparison to ensure that the predicted flow behavior reasonably represents the physical condition.

In the experimental investigation, scaled tanker models were placed on a moving platform. The platform setup converted rotational motion into linear reciprocating motion, which represented acceleration induced sloshing in road tankers. The sloshing force was measured using a load cell and the liquid movement was observed during the test. In the numerical part, the same tanker geometries and operating conditions were modelled to visualize the liquid free surface, flow contours and force response.

The methodology was carried out in a step by step sequence. At first, the tanker geometries were selected and designed. Then the unbaffled cases were analyzed to understand the basic sloshing response of each tanker shape. After that, perforated longitudinal and transverse

baffles were introduced as mitigation techniques. Finally, the numerical and experimental results were compared to evaluate the effect of tanker geometry and baffle arrangement on sloshing reduction.

2.1.1 Flow Chart of Research Methodology



Figure 2.1: Overall workflow of the research methodology

2.2 Experimental Methodology

The experimental investigation was conducted to observe the actual sloshing behavior of liquid inside the tanker under controlled back and forth excitation and to validate the numerical model. In the experimental study, only the rectangular tanker was considered. Both unbaffled and baffled conditions were examined to evaluate the effect of internal baffles on sloshing

behavior. In the baffled cases, perforated longitudinal and transverse baffles were used as sloshing mitigation devices.

A mechanical setup was developed to generate reciprocating motion of the tanker. The excitation was produced using a motor wheel connecting rod mechanism, which converted rotary motion into linear back and forth motion. This arrangement was selected to simulate the motion responsible for sloshing during transportation. During the experiment, the liquid flow pattern and contour development were visually observed, while the force generated due to sloshing was measured using a load cell. The experimental results were later compared with the corresponding numerical results of the rectangular tanker for validation.

2.2.1 Experimental Tanker Model

The experimental study was carried out using a rectangular tanker model with volume same as the volume of real life circular tanker when scaled as 1:20. The tank was fabricated with suitable transparent material (acrylic sheet) so that the liquid motion inside the tank could be observed clearly during the test. The dimensions of the tank were selected according to the design requirements of the study and the length was 800 mm and width and height was both 200 mm. In addition to the unbaffled model, baffled models were also prepared by incorporating perforated longitudinal and transverse baffles inside the tank. The dimensions and positions of the baffles were maintained according to the selected test conditions.

2.2.2 Experimental Setup

The experimental setup consisted of the rectangular tanker model, a supporting frame, a movable platform, a motor, a rotating wheel, a connecting rod mechanism and a load cell. The purpose of the setup was to generate controlled reciprocating motion of the tank and to measure the force response created by sloshing. The overall arrangement was designed in such a way that the tanker could move in a back and forth direction while maintaining stability during operation.



Figure 2.2: Experiment setup of present study

Table 2.1: Main components used in the experimental setup

Components	Description	Function in the experiment
Rectangular tanker model	Acrylic tank, 800 mm × 200 mm × 200 mm	Hold the liquid
Working fluid	Water, 50% filling level	Represents liquid cargo
AC geared motor	Speed controlled motor	Provides rotary motion
Rotating wheel	Attached to motor shaft	Transfer motor rotation
Connecting rod	Linkage	Converts rotary motion into linear motion
Moving platform	Tank supporting platform	Moves the tank back and forth
Load cell	10 kg strain gauge sensor	Measures sloshing force
HX711 amplifier	24 bit amplifier module	Amplifies load cell signal
Arduino Uno	Microcontroller board	Reads force data
Baffles	Longitudinal and perforated baffles	Reduces liquid motion



(a)



(b)



(c)



(d)

Figure 2.3: Main Components Used in Experimental Setup (a)AC Gear Motor with Controller, (b) Arduino UNO, (c) Load Cell Module and (d) HX711 Amplifier

2.2.3 Excitation Method

A motor wheel connecting rod mechanism was used to generate the required reciprocating motion of the tanker. In this mechanism, the rotary motion of the motor was transferred to a wheel and the rotation of the wheel was converted into linear back and forth motion through a connecting rod. This arrangement was used to simulate the oscillatory motion responsible for liquid sloshing inside the tank.

The rotational speed of the motor was maintained at 20 rpm, which corresponds to an excitation frequency of 0.333 Hz. The corresponding angular frequency was calculated as 2.09 rad/s. Based on the measured peak acceleration of 1.09 m/s^2. Therefore, the excitation system produced a controlled reciprocating motion with a frequency of 0.333 Hz, which was used throughout the experimental analysis.

2.2.4 Instrumentation Setup and Experimental Procedure

A load cell was used as the primary instrument to measure the changing load exerted due to the liquid sloshing. It was attached to the rectangular tanker at the side wall of the tank, so as to directly measure the effect of pressure due to the oscillating liquid on the wall of the tank.

Upon the back and forth excitation of the tank, a time dependent sloshing motion was induced in the liquid of the tank, which resulted in fluctuations in the forces acting on the wall of the tank. These fluctuations were measured by the load cell attached to the tanker and were converted to electrical signals. These electrical signals were then further amplified using an amplification device (HX711) attached to the load cell. The amplified signal was fed to a reading system which acted as a unit for reading or acquiring data from the load cell. The readings from the load cell, after being amplified and processed, were received by this unit and were further analyzed during the test to determine the force values.

The force data collected was used to present the sloshing induced changing response of the tanker due to the applied excitation. It was also utilized to analyze the effect of the baffles in reducing the force and finally used to compare the experimental results with the numerical force predictions.

Other equipment and gadgets that were used during the experiment includes an excitation assembly consisting of the supporting frame and the force reading unit. The experiment was conducted using a transparent tank which enabled the analysis of the flow pattern and the development of the contour inside the tank. could be compared with the simulation results.

At the beginning of each experiment, the tank was filled with water so as to fill the test tank to a depth of 50%. It is then mounted on the oscillation machine.

Then, the machine is activated so that the tank begins to oscillate. Once the motion began, the motion of the water was monitored while it is observed whether the generated waves were appropriate. The force exerted on the tank is measured using the pressure sensor attached to it. After the procedure described above has been checked visually that the water level inside the tank is suitable, these operations are repeated for every state to be compared with each other.

Table 2.2: Specifications of force measuring sensor (load cell)

Components	Specification
Force sensor	Strain gauge load cell
Rated capacity	10 kg, approximately 98.1N
Signal conditioning unit	HX711 load cell amplifier module
ADC resolution	24 bit
Function in experiment	Amplified and digitized the load cell signal for sloshing force measurement

2.2.5 Calibration Measurement and Uncertainty

In the beginning, the load cell was calibrated with standard weights to make sure the readings obtained reflected the real forces applied. The zero reading was adjusted before each test, as this eliminated any bias present due to the initial setting of the device. Calibration assisted in converting the electric output from the load cell into force measurements during the sloshing experiment.

The slight variations in the measurement of force during the experiment were expected due to the sensitivity of the sensor, vibrations of the mobile platform, positioning of the tank and some error in manual adjustment. This uncertainty did not play a significant role because the consistent approach was used throughout the experiments with both configurations of the tank.

2.2.6 Safety Precautions

The following safety precautions were maintained during the experimental work:

1.The tanker model was properly fixed on the moving platform before starting each test to prevent unwanted movement or slipping.

2.The connecting rod, rotating wheel and moving platform were checked before operation to make sure that all parts were properly attached and aligned.

3.Hands, tools and loose objects were kept away from the moving mechanism while the motor was running.

4.The motor, load cell, HX711 amplifier and data reading connections were inspected before each test to avoid loose wiring or signal interruption.

5.Water leakage was checked carefully before and during the experiment to prevent contact between water and electrical components.

6.The experiment was operated at the selected controlled speed to avoid excessive vibration, sudden impact or damage to the setup.

7.The setup was monitored throughout the test so that the motor could be stopped immediately if abnormal vibration, leakage or mechanical looseness was observed.

2.3 Numerical Methodology

Numerical simulation was performed in ANSYS Fluent to analyze liquid sloshing inside partially filled tanker models. A multiphase flow approach was used to capture the air water interface and observe the free surface variation during excitation. The simulations were conducted for the selected tanker geometries with both unbaffled and baffled configurations. The numerical results were later compared with the experimental observations for validation of the rectangular tanker model.

2.3.1 Software and Computational Platform

The numerical analysis was conducted using ANSYS Fluent. The tanker geometries were modelled according to the selected design dimensions and the computational domains were prepared for rectangular, capsule shaped and cylindrical tankers. The simulations were performed for both unbaffled and baffled configurations to evaluate the effect of perforated longitudinal and transverse baffles.

The computational work was carried out on a personal laptop with the following specifications:

Table 2.3: Software and Computational Platform Specifications Used for Numerical Simulation

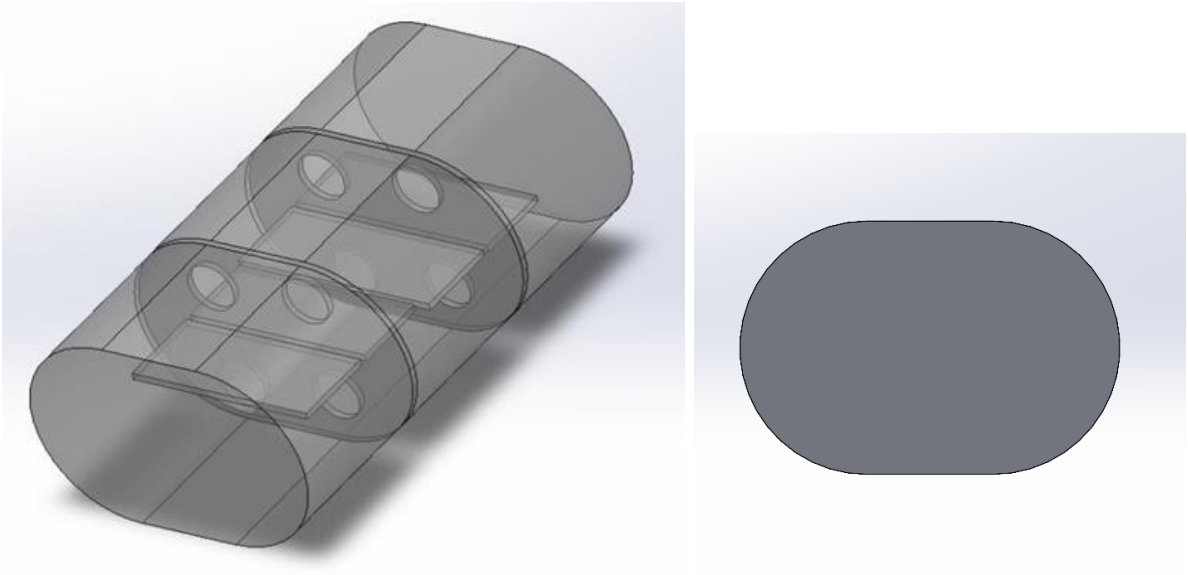
Item	Specifications
Simulation software	ANSYS fluent
Solver type	Pressure based transient solver
Multiphase model	Volume of fluid(VOF)
Working fluids	Water and air
Processor	Intel Core i5-1135G7
RAM	8GB
Operating system	Windows 10

2.3.2 Geometry Development

The numerical models were developed for three different tanker geometries: capsule, cylindrical and rectangular. The dimensions of the tanks were selected to allow a fair comparison among the different shapes under the same excitation condition. In addition to the unbaffled models, baffled configurations were also created by incorporating longitudinal and transverse baffles inside the tank. The baffle dimensions and positions were defined according to the selected design conditions. The detailed geometric dimensions of the tankers and baffles are presented in table:

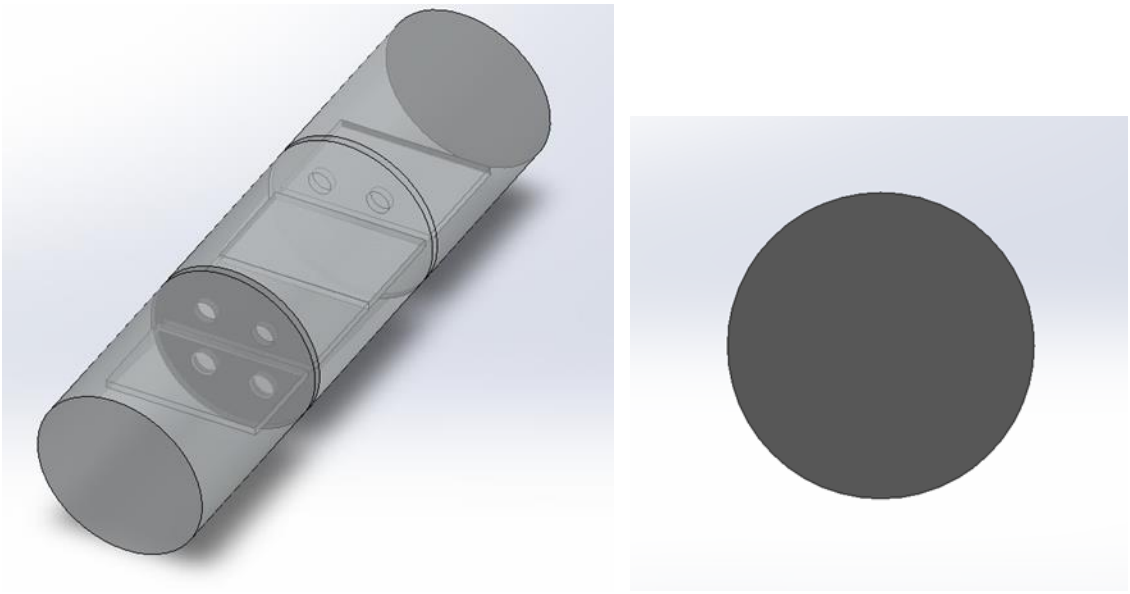
Table 2.4: Dimensions of tankers of different geometries

Capsule	Cylindrical	Rectangular
Length: 800 mm	Length: 800 mm	Length: 800 mm
Length of Flat portion:150mm	Diameter: 220 mm	Width: 200 mm
Radius:150 mm		Height: 200 mm



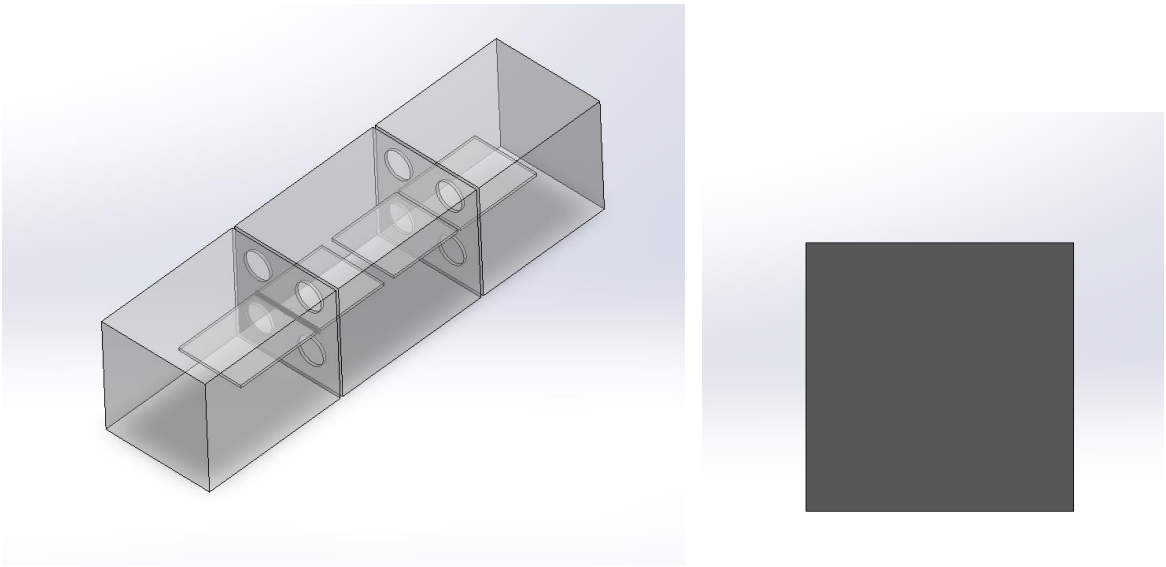
(a) (b)

Figure 2.4: Three dimensional CAD model of capsule shaped tanker, (a) Isometric view and (b) Cross sectional view



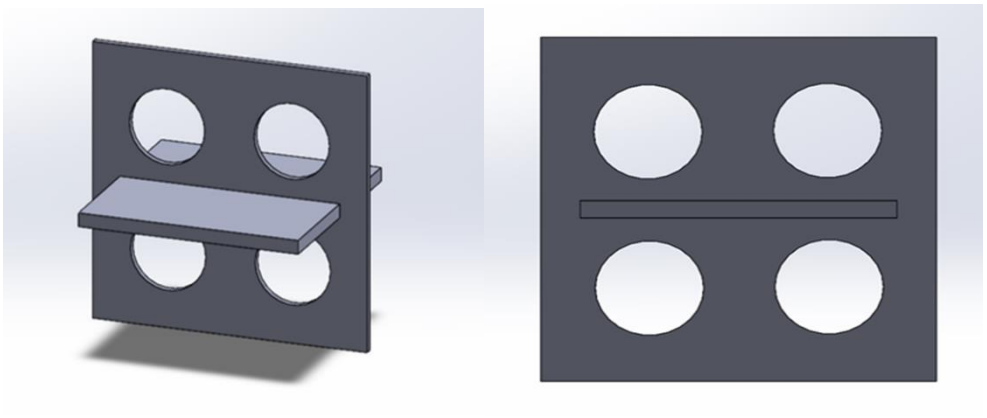
(a) (b)

Figure 2.5: Three dimensional CAD model of circular shaped tanker, (a) Isometric view and (b) Cross sectional view



(a) (b)

Figure 2.6: Three dimensional CAD model of rectangular shaped tanker, (a) Isometric view and (b) Cross sectional view



(a) (b)

Figure 2.7: Three dimensional CAD model of Baffle, (a) Isometric view and (b) Cross sectional view

Here are the CAD models of our concerned tankers: capsule, cylindrical and rectangular are shown respectively in Figure 2.4, Figure 2.5 and Figure 2.6 and the CAD model of baffle is shown in Figure 2.7.

2.3.3 Governing Equations

The numerical simulation was carried out in ANSYS Fluent using the transient Volume of Fluid (VOF) model. The VOF method was used to capture the free surface motion between water and air inside the partially filled tanker. The flow was considered incompressible and unsteady. The governing equations used in the simulation are given below.

The continuity equation for incompressible flow is:

$$\nabla . u = 0 \qquad \textbf{2.1}$$

where, u is the velocity vector.

The transient momentum equation is:

$$\frac{\partial(\rho\vec{u})}{\partial t} + \nabla \cdot (\rho\vec{u}\vec{u}) = -\nabla p + \nabla \cdot [\mu(\nabla u + \nabla u^T)] + \rho g + S_m \qquad \textbf{2.2}$$

where, ρ is the mixture density, p is the pressure, μ is the mixture dynamic viscosity, g is the gravitational acceleration, and S_m is the momentum source term due to the imposed excitation.

The volume fraction equation for the water phase is:

$$\frac{\partial \alpha_w}{\partial t} + \nabla \cdot (\alpha_w u) = 0 \qquad \textbf{2.3}$$

where, α_w is the volume fraction of water.

For the air water system, the summation of volume fractions is:

$$\alpha_w + \alpha_a = 1 \qquad \textbf{2.4}$$

where, α_a is the volume fraction of air. A computational cell contains only water when $\alpha_w = 1$, only air when $\alpha_w = 0$ and the air water interface when $0 < \alpha_w < 1$.

The mixture density is calculated as:

$$\rho = \alpha_a \rho_a + \alpha_w \rho_w \qquad \textbf{2.5}$$

The mixture dynamic viscosity is calculated as:

$$\mu = \alpha_a \mu_a + \alpha_w \mu_w \qquad \textbf{2.6}$$

where ρ_w and ρ_a are the densities of water and air respectively and μ_w and μ_a are their corresponding dynamic viscosities.

The reciprocating motion of the tanker was represented by a time dependent acceleration in the longitudinal direction. The acceleration input was defined as:

$$a_x(t) = 1.09 \sin(2.09t) \qquad \textbf{2.7}$$

where $a_x(t)$ is the time dependent acceleration, 1.09 m/s² is the measured peak acceleration, and 2.09 rad/s is the angular frequency corresponding to the motor speed of 20 rpm.

2.3.4 Mesh Generation and Mesh Independence Study

For the discretization of the computational domain, tetrahedral elements were used in the pre processing stage. This kind of element selection was necessary for their suitability in the meshing process of complex geometries. More importantly, tetrahedral elements were considered appropriate due to their ability to conform to the different shapes of tankers considered in the current analysis. Careful attention was paid to the water air interface region because of the importance of free surface displacement analysis in the context of sloshing dynamics. The appropriate mesh was determined taking into account numerical accuracy and computational efficiency.

A mesh convergence study was performed in order to minimize any possible dependency on the size of mesh. For that reason, sloshing simulations were conducted using different meshes based on the density of tetrahedral elements. The results from each mesh were compared in terms of some selected output variables. Free surface dynamics and sloshing forces response were considered as some representative output parameters. After certain mesh density levels, any further decrease in mesh size caused only minor changes in the results. As a result of this, the mesh size of 5 mm is determined. This observation is presented in the following table:

Table 2.5: Variation of parameter with respect to mesh size

Mesh size	Parameter (Force)
10 mm	15.28N
5 mm	14.6N
3 mm	14.66N

2.3.5 Initial and Boundary Conditions

The tank walls and baffles were treated as rigid no slip walls. The tank was initially filled with water up to the selected fill level (50%), while the remaining portion of the tank was occupied by air. The initial liquid surface was defined before the start of the transient simulation. Gravity was applied in the vertical direction to represent the actual liquid behavior inside the tank.

The initial velocity of the liquid was taken as zero and the motion of the tank was then introduced through the prescribed excitation condition. These initial and boundary conditions were used consistently for all the tanker geometries in order to maintain uniformity in comparison.

2.3.6 Solver Settings and Convergence Criteria

A pressure based laminar solver was used for the simulation. The Volume of Fluid model was selected to capture the air water interface. The spatial discretization schemes, time step size and convergence criteria were selected to obtain stable and reliable results.

The convergence of each time step was monitored using residual values and force response. A sufficiently small time step was used to capture the free surface motion and force fluctuation during sloshing.

Table 2.6: Solver settings used in ANSYS Fluent

Setting	Selected Option
Solver	Pressure based
Time formulation	Transient
Multiphase Model	VOF
Fluids	Water and air
Time step size	0.01s
Number of time step	600
Residual convergence criterion	10^{-4}

2.3.7 Model Validation

To validate the accuracy of our CFD setup in ANSYS Fluent, we replicated a bench mark case from the paper titled *"Liquid Sloshing in a Two-Dimensional Rectangular Tank: A Numerical Investigation with a T-Shaped Baffle."* This validation step was essential to ensure that our modeling approach could reliably capture the physics of sloshing before applying it to more complex 3D geometries. The validation was conducted for the unbaffled tank case with the following parameters: a two dimensional rectangular tank, 50% fill level, turbulent flow regime and a prescribed excitation angle of $\theta = 8°$. The Volume of Fluid (VOF) method was used to capture the free surface motion of the liquid. The tank motion was modeled using a dynamic body force to simulate the imposed excitation.

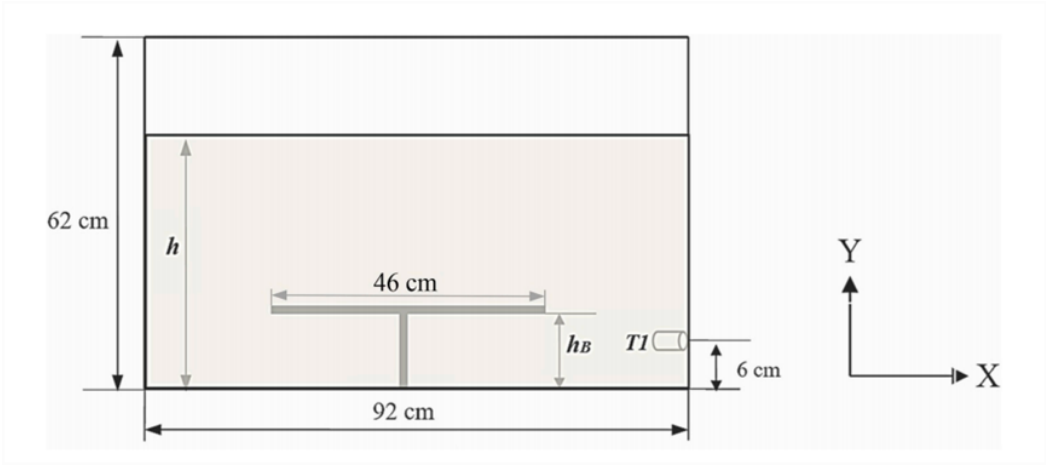


Figure 2.8: Setup used in computational study of Unal et al.'s **[16]**

Table 2.7: Variation of non dimensional pressure(P/ρgh) with respect to wt (50% filling depth,unbaffled,turbulent,$\theta_0 = 8°$)

ωt (rad)	P/ρgh Ünal et al. [16]	P/ρgh [Present Setup]	P (Pa) Ünal et al. [16]	P (Pa) [Present Setup]
1.00	0.59	0.63	1790.65	1914.96
2.00	0.55	0.52	1669.25	1577
3.00	0.66	0.64	2003.10	1943.53
4.00	1.12	1.03	3399.20	3116
5.00	1.20	1.14	3642	3464.44
6.00	0.97	0.96	2943.95	2912
7.00	0.55	0.55	1669.25	1670.8
8.00	0.42	0.39	1274.70	1182
9.00	0.87	0.72	2640.45	2195.5
10.00	1.02	1.00	3095.70	3035
11.00	1.03	1.08	3126.05	3303.6
12.00	0.98	0.95	2974.30	2875.47
13.00	0.78	0.70	2367.30	2111.37
14.00	0.54	0.50	1638.90	1517

For comparison, we measured the non dimensional pressure at a specified monitoring point on the tank wall, consistent with the reference study. We then plotted variation of non dimensional pressure with ωt to visualize the pressure response over a full oscillation cycle. The resulting waveform was directly compared with the pressure plot provided in the reference paper. In addition to visual comparison, key numerical values such as peak pressure and oscillation

period were extracted and compared. The difference between our simulation results and those reported in the literature remained within a reasonable margin of error.

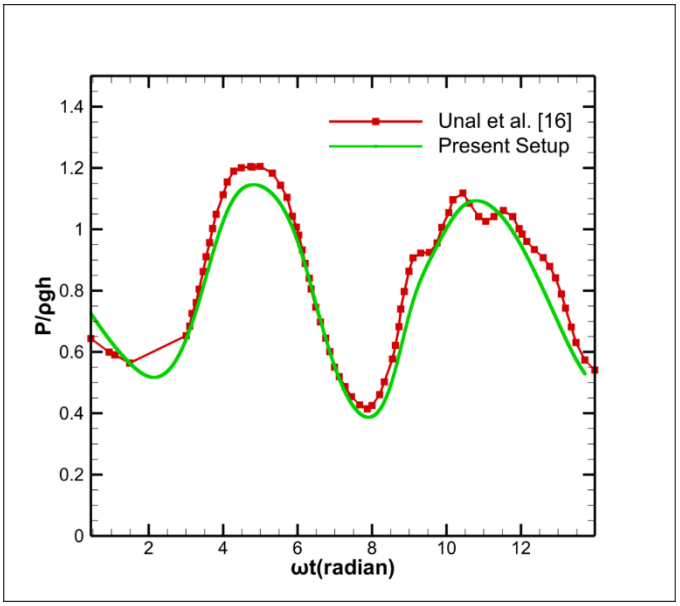


Figure 2.9: Comparison of static pressure with present setup and [16]'s result for 50% filling depth, unbaffled, turbulent,$\theta_0 = 8°$

This successful validation confirmed the reliability of our simulation settings, including turbulence modeling, mesh resolution, time step size and boundary condition implementation. With confidence in the model accuracy, we proceeded to apply the same approach to simulate sloshing in cylindrical and elliptical tanks with various baffle configurations.

2.4 Use of Engineering Tools, Data Collection & Processing

2.4.1 Mesh Structure and Sensor Logs of Load Cell

Figure 2.8 shows the computational mesh generated for the tanker model used in the numerical sloshing simulation. A uniform mesh distribution was maintained throughout the computational domain to keep the discretization consistent in all regions of the tank. This mesh was used to represent the internal air water domain and to capture the transient movement of the free surface during excitation. Before running the simulation, the mesh quality was checked to ensure that the model was suitable for stable and reliable numerical calculation.

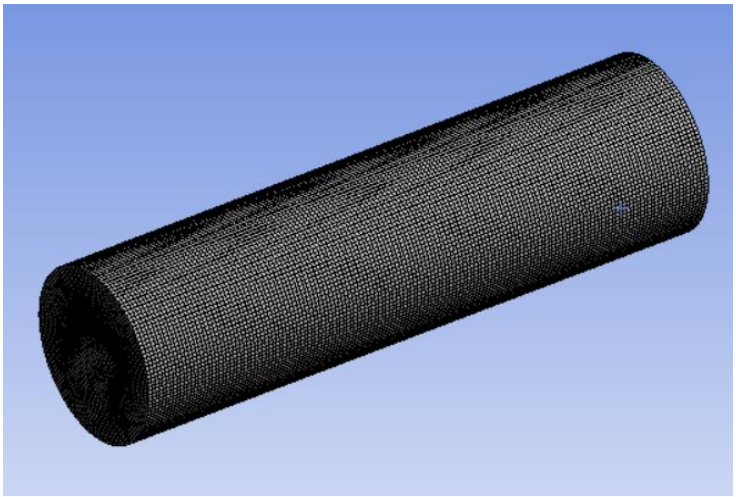


Figure 2.10: Computational Mesh Generated for the Tanker Model in ANSYS Fluent

In the process of experimentation, the force created due to the movement of the liquid was measured using load cells and HX711 amplifier module and Arduino Uno. The results were recorded in the form of sensor logs wherein each measurement was related to the force created by the liquid at that point in time. These sensor logs provided information about the effect of time on the sloshing of the liquid and also the role played by the baffle in this process.

Sensor logs from two different conditions were prepared. In the first case, sensor logs were collected without the presence of a baffle and this is when the liquid moved freely in the rectangular tank. In the second case, the sensor logs were obtained under the condition where the internal baffle was present inside the tank and acted as an impediment to the free movement of the liquid.

Table 2.8: Sensor log for the unbaffled rectangular tanker model

Time(s)	Force(N)
0	0
0.12	0.6
0.15	1.7168
0.21	7.8037
0.25	9.33
0.45	13.61
0.6	17.66
0.8	21.337
1.1	24
1.3	18
1.45	12.2
1.67	7.16
1.9	5.22
2	3.17
2.3	1.629
2.47	1.39

Table 2.9: Sensor log for the baffled rectangular tanker model

Time(s)	Force(N)
0	0
0.12	0.3
0.19	0.6
0.25	7.4
0.3	10.3
0.45	11.9
0.6	10.5
0.81	13.4
0.93	14.02
1.1	15.52
1.3	13.75
1.42	11.11
1.64	10.811
1.8	7.93
1.9	6.77
1.98	2.699
2	2.113
2.1	1.87
2.3	1.8
2.47	1.81

The comparison between the two tables shows the difference in force response between the unbaffled and baffled cases. In the unbaffled condition, the liquid motion was more unrestricted, which generally produced larger force variation. In the baffled case, the internal baffle interrupted the liquid movement and helped reduce sudden force changes. Therefore, the sensor logs provided direct experimental evidence of the influence of baffles on sloshing control.

2.4.2 Use of Engineering Tools and Justification of Tool Selection

ANSYS Fluent was selected for the numerical part of this study because it is suitable for simulating sloshing problems involving water and air. Since the main focus of the work was to observe the movement of the liquid free surface, the Volume of Fluid (VOF) model in Fluent was appropriate for capturing the air water interface, wave formation and force response inside the tanker. Fluent also provides convenient options for defining material properties, boundary conditions, transient solver settings, mesh checking and post processing of contours and force data, which made it useful for comparing different tanker geometries and baffle arrangements.

Compared with OpenFOAM, ANSYS Fluent was more convenient for this study because it has an integrated graphical interface and requires less manual coding during model setup. This helped reduce setup complexity and allowed the simulation results, such as volume fraction contours and force plots to be obtained more efficiently.

For experimental data reading, an Arduino Uno was used with the HX711 load cell amplifier module. The HX711 amplified and converted the low voltage signal from the 10 kg load cell into digital data and the Arduino Uno was used to read and transfer the force data for recording. This arrangement was selected because it is simple, low cost, compatible with load cell measurement and suitable for collecting time dependent force readings during the sloshing experiment.

Chapter 3

Result and Discussion

Sloshing can be described as the combination of multiple surface waves. The behavior of these waves are governed by different forces, like, in non rotating motion, gravity and surface tension primarily dominates whereas in rotating motion centrifugal and coriolis forces come into play along with those. In the present study, the tanks was subjected sinusoidal acceleration along the length of the container. The variation of pressure and force on the wall of the tank has been investigated for three different tank models along with the effect of the baffle in each model.

3.1 Results of Numerical Investigation on the Effect of Tank Cross Section

Three cross sections of tank has been numerically studied; Rectangular, Circular and Capsular. The geometry of the tanks plays vital role in determining the sloshing effect whether the tank has more sharp edge or curved edge or mixed geometry. The volume of each tank was kept same and the pressure and the force was analysed at the same location for all of the tanks.

3.1.1 Pressure Comparison Among Tanks

The pressure measurement in this section is based on hydrostatic pressure which is mainly governed by the shift in center of mass during sloshing. The pressure variation due to the tank geometry are illustrated for baffled and unbaffled tanks at a height of H=0.05m from the bottom wall.

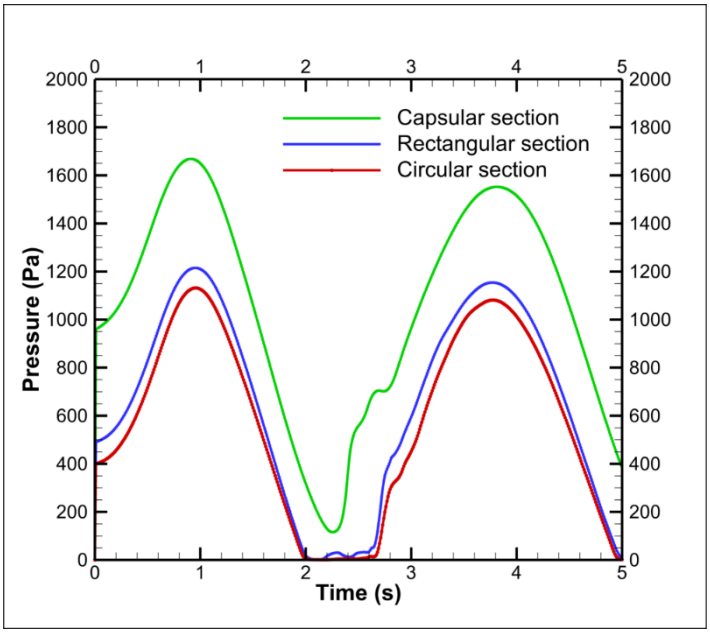


Figure 3.1: Pressure variation of different cross sectional tanks without baffle

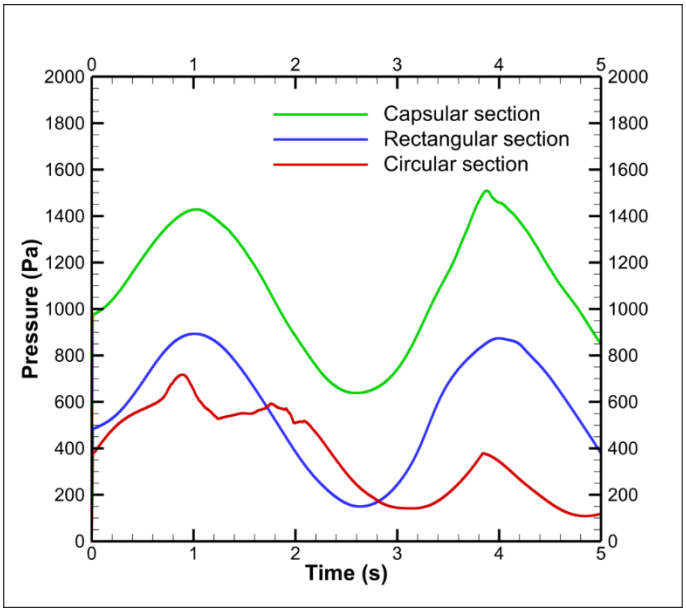


Figure 3.2: Pressure variation of different cross sectional tanks with baffle

3.1.2 Force Comparison Among Tanks

The sloshing force is the force exerted by the moving fluid on the walls of the container which is influenced by the excitation frequency, liquid depth, tank geometry etc. The variation of the force measured in this segment is based on the variation in the cross sectional geometry of the tanks.

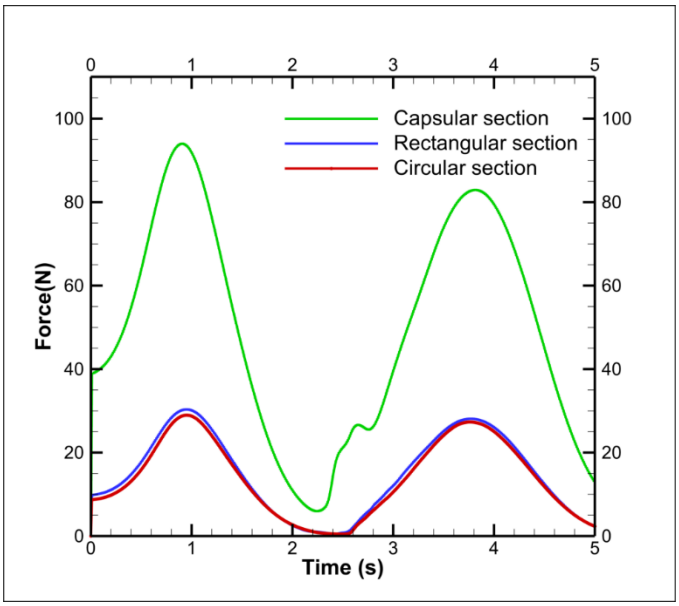


Figure 3.3: Force variation of different cross sectiona tanks without baffle

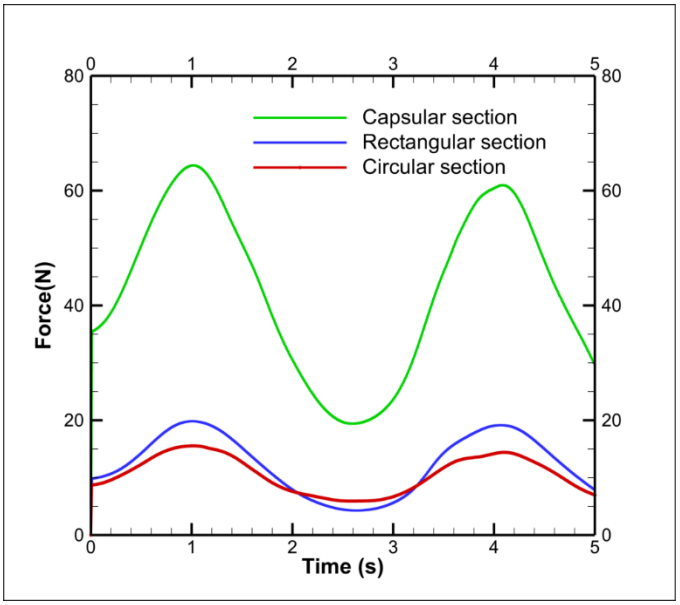


Figure 3.4: Force variation of different cross sectional tanks with baffle

3.2 Analysis and interpretation of the effect of geometry

In Figure 3.1 and Figure 3.2, the pressure distributions corresponding to tanks with different cross sectional geometries are numerically evaluated at a height of H=0.05m from the tank bottom. Figure 3.3 and Figure 3.4 represent the variation in force due the differences in the geometry of the tank cross section. The results indicates that the capsular shaped tank yields the highest pressure at that height and the highest force at the same wall, whereas the circular one exhibits the lowest values of pressure and force at the identical locations.

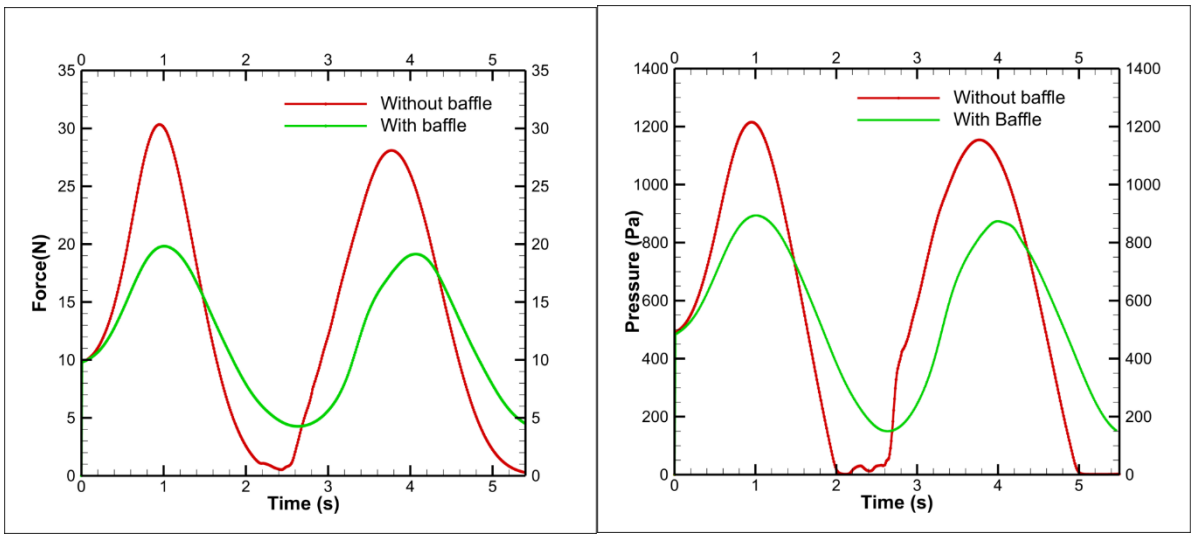
The main reason behind capsular tanks to show the higher values of pressure and force is attributed to the comparatively greater height dimension of the tank(300mm) than the other two tanks(200 and 220mm), thereby increasing the hydrostatic pressure at the specified height for same volume of tank. Moreover, in the capsular shape, the combination of the flat side and the curve wall creates a hybrid effect which leads to amplified sloshing effect. The rectangular shaped tank giving more pressure and force values than the circular section is likely influenced by the presence of sharp corners, which enhances wave reflection and fluid confinement, leading to localized pressure amplification in the rectangular geometry. The circular shape distributes wave energy smoothly, hence, the lowest values for pressure and forces.

3.3 Results of Numerical Investigation on the Effect of Baffle

A combination of perforated vertical baffle and horizontal baffle has been used in the study. Total two baffles are used which are located symmetrically at a distance of 260mm from both left and right tank wall. Both qualitative and quantitative force and pressure variations based on the presence of the baffle is showed in this section for all three tanks separately.

3.3.1 Tank of Rectangular Cross Section

The Figure 3.5 shows the quantitative comparison of the force and pressure on the wall of the tank between unbaffled and baffled rectangular tank.



(a) (b)

Figure 3.5: The(a) Force and (b)Pressure variation of the rectangular cross sectional tank

The free surface elevation images was also compared in Figure 3.6 from the software post processing. The images from the left side represents the unbaffled tank and the right images are from the baffled tank analysis. The result are showed at three different time steps for comparison.

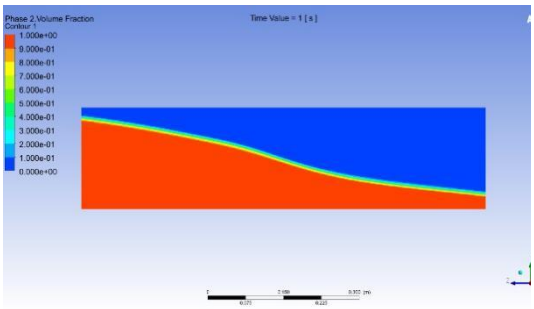
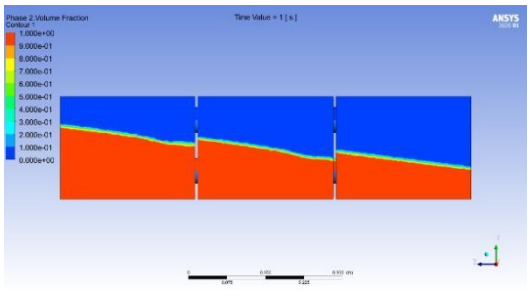
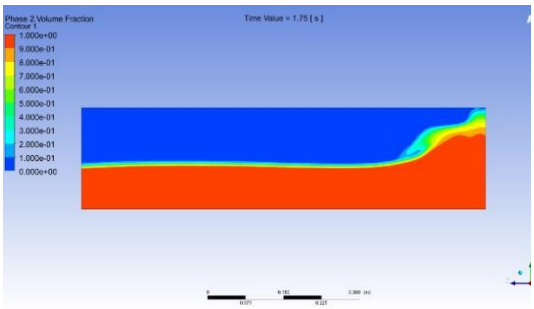
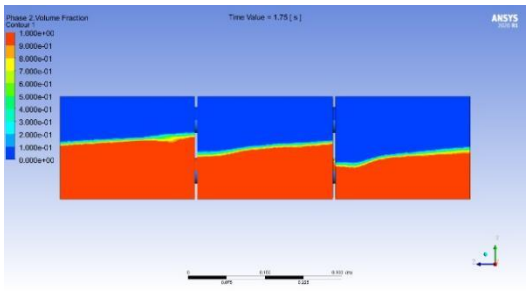
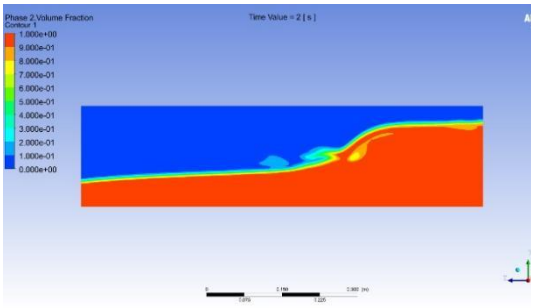
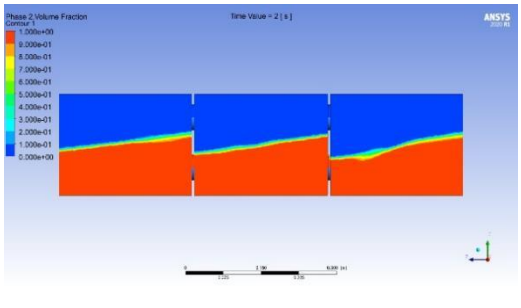
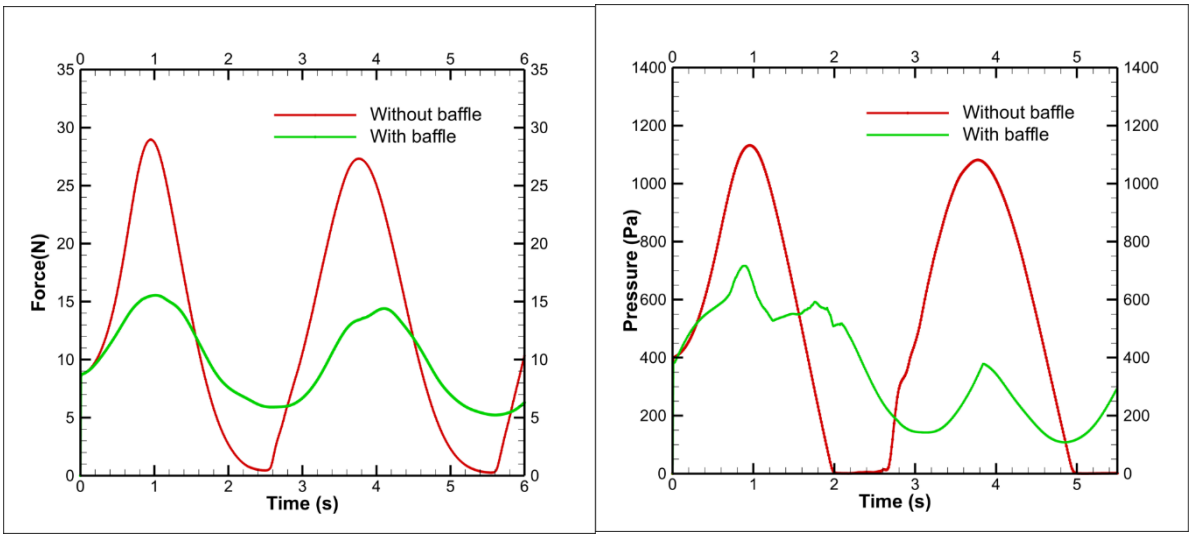
Time	Without Baffle	With Baffle
T=1s		
T=1.75s		
T=2s		

Figure 3.6: Comparison of free surface elevation between rectangular tank with and without baffle

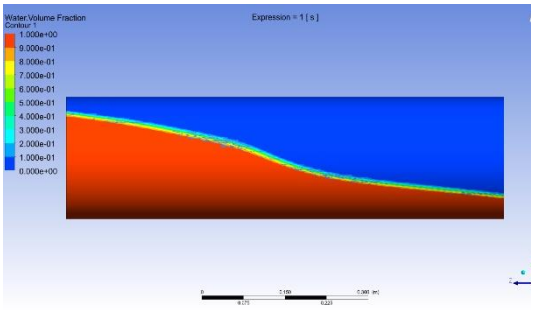
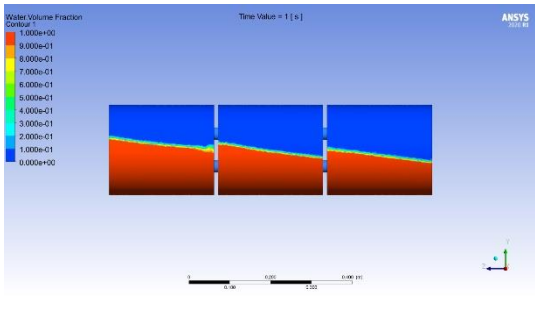
3.3.2 Tank of Circular Cross Section

The numerical comparison between the baffled and unbaffled tank with circular cross section is illustrated in this section. Figure 3.7 and Figure 3.8 shows the result of quantitative and qualitative analysis respectively.

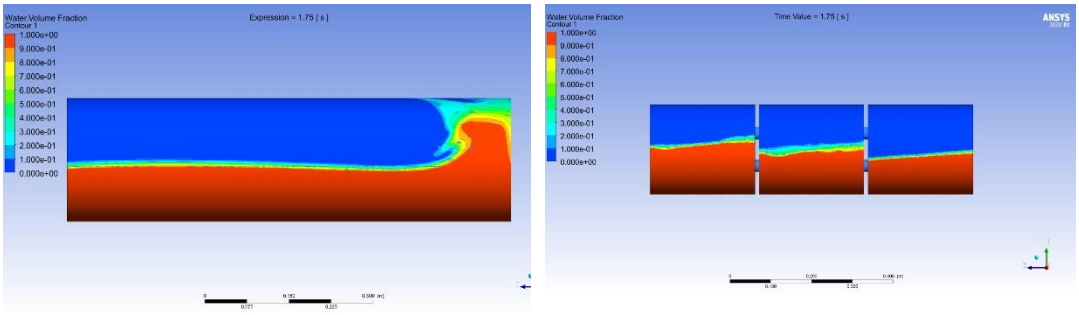


(a) (b)

Figure 3.7: The(a) force and (b)Pressure variation of the circular cross section tank

Time	**Without Baffle**	**With Baffle**
T=1s		

T=1.75s



T=2s

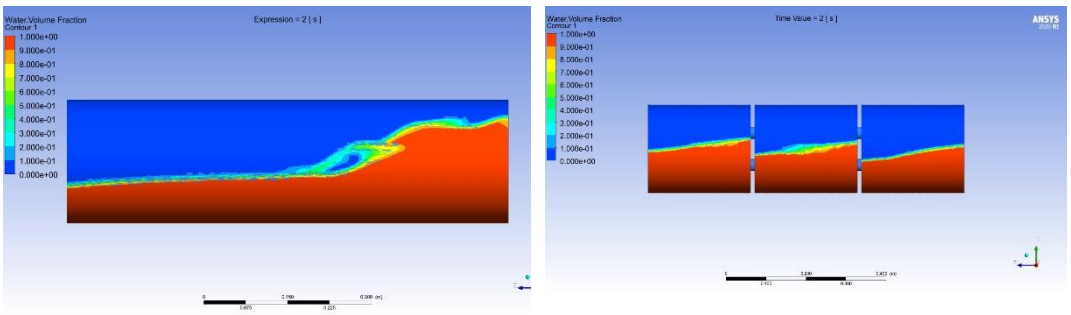
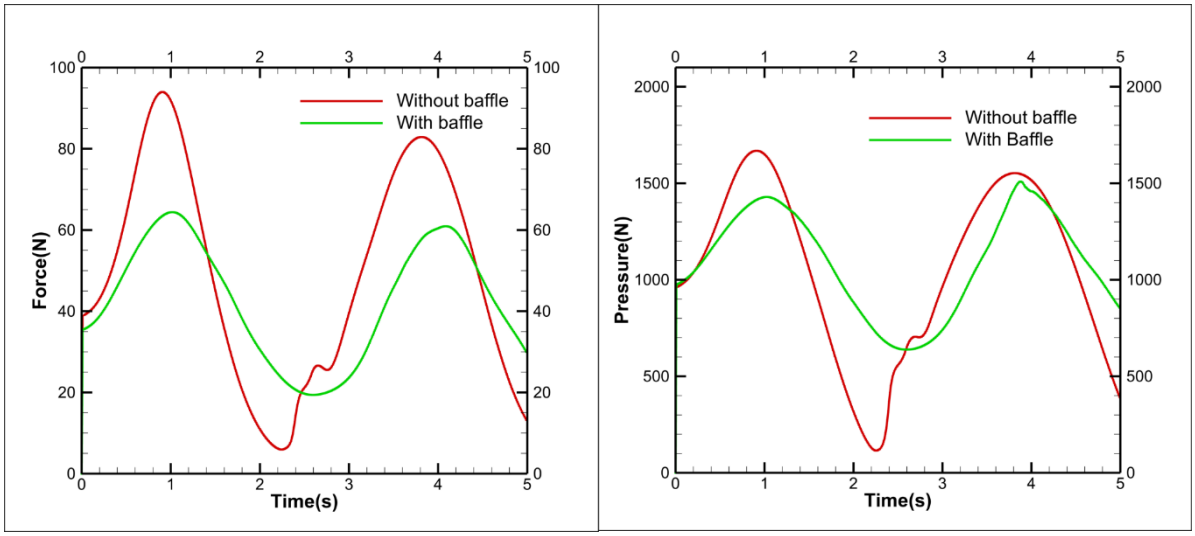


Figure 3.8: Comparison of free surface elevation between circular tank with and without baffle

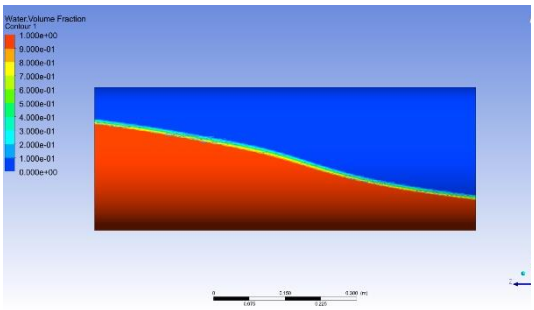
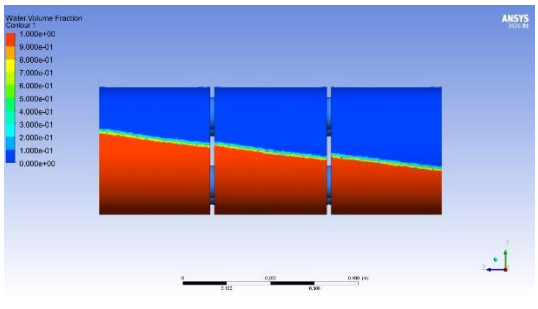
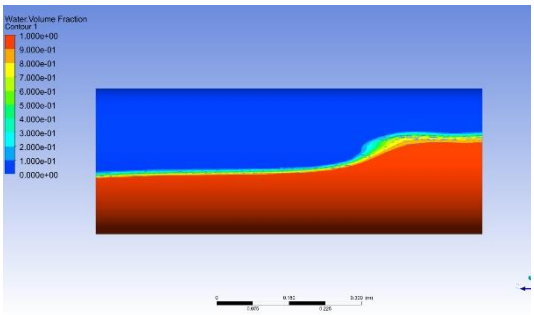
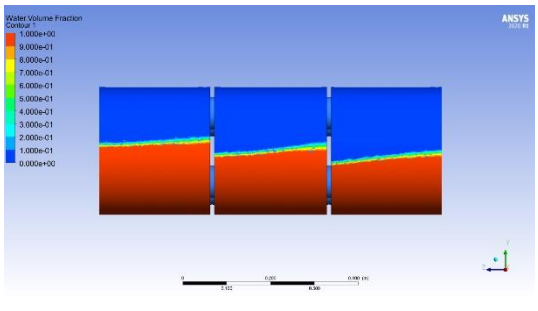
3.3.3 Tank of Capsular Cross Section

The comparison demonstrated in Figure 3.9 and Figure 3.10 reveals the variation in pressure and force at the same place on the tank wall with capsular cross section due to the effect of baffles. The analyses performed in this section are also numerical quantitative and qualitative.



(a) (b)

Figure 3.9: The(a) force and (b)Pressure variation of the capsular cross section tank

Time	**Without Baffle**	**With Baffle**
T=1s		
T=1.75s		

T=2s

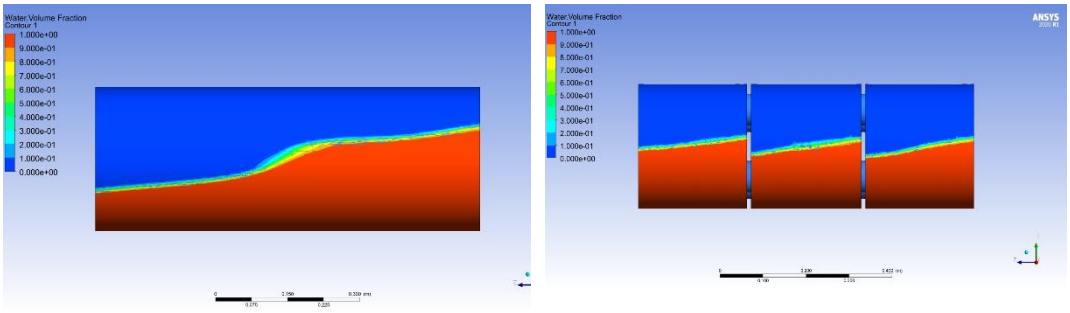


Figure 3.10: Comparison of free surface elevation between capsular tank with and without baffle

3.4 Result of experimental investigation on the effect of baffle

Figure 3.11 shows the effect of baffle on force and pressure in a rectangular tank during the experimental analysis. Also, in Figure 3.12, the still images taken during experimental study reveals the effect of baffle on free surface elevation of the fluid.

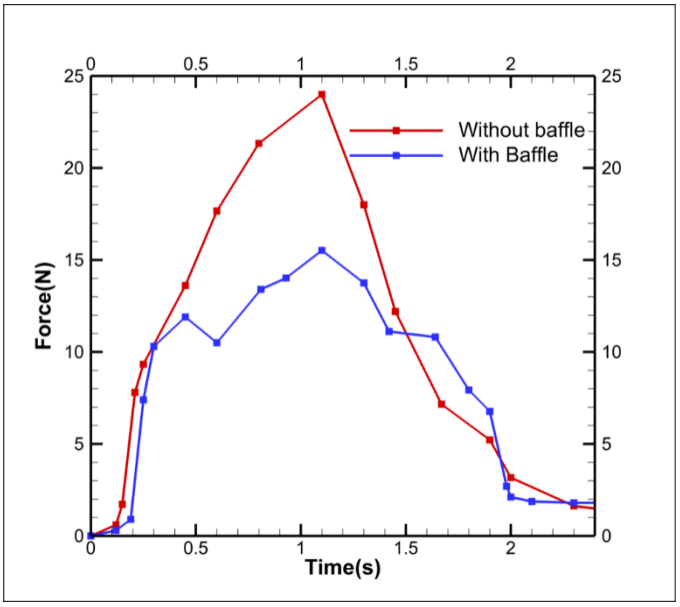


Figure 3.11: Variation of force on the wall of rectangular tank in experimental analysis

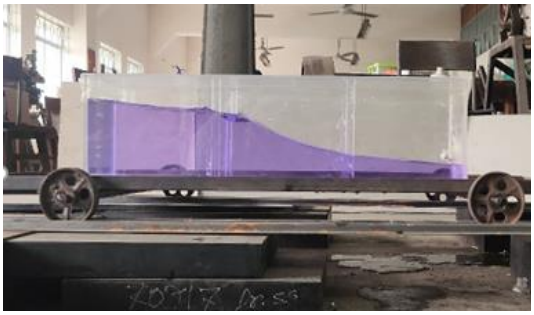
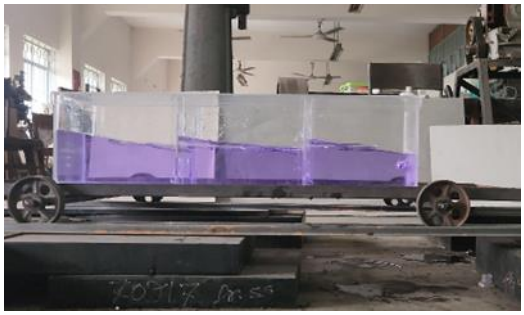
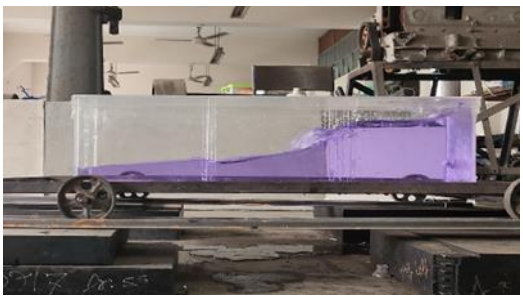
Time	Without Baffle	With Baffle
T=1s		
T=1.75s		
T=2s		

Figure 3.12: Comparison of free surface elevation between unbaffled and baffled rectangular tank in experimental study

3.5 Analysis and interpretation of the effect of baffle

Figure 3.5, Figure 3.7 and Figure 3.9 show the quantitative results of the numerical comparison between baffled and unbaffled tanks of different cross sections. On the other hand Figure 3.6, Figure 3.8 and Figure 3.10, represents the qualitative comparison for these in numerical study.

The force and pressure is reduced for all three tanks due to addition of baffle. Though the degree to which these are reduced by the baffle is less in case of the capsular tank. In Figure 3.10 also, it is noticeable that the free surface elevation of baffled tank is not much different

from that of the unbaffled tank. It proves that the hybrid effect attained by the complexity of the geometry in capsular section dominates over the positive effect caused by the baffle.

The other two tanks show a prominent effect of baffle in force and pressure leading the circular cross section to have the utmost effect of baffle. The free surface elevation presented for these two geometries of tank in Figure 3.6 and Figure 3.8 shows the advantage of baffle in mitigating the free surface elevation of the fluid, specially in case of the highest elevation shown in t=1.75s in both of the cases.

In section 3.4, we can see the experimental quantitative and qualitative analysis results which also indicate that the baffle indeed reduces the peak pressure and the force values in the rectangular tank. In Figure 3.12, it is illustrated that the wave displacement is decreased due to the baffle.

The main reason for a baffle to be able to mitigate the sloshing effect is because of the compartmentalization of the tank. Due to the vertical member of the baffle, the whole tank is divided into three separate compartments lessening the height of the free surface elevation and reducing the pressure. Moreover, impact from the sloshing reduces as each compartment contains only partial mass of the total fluid, hence, fluid in each compartment generates lower force on the baffle. So, it has less overall effect on the outermost wall than the effect of the whole mass of fluid on the tank wall.

The perforation used in the vertical member of the baffle allows the fluid to pass through it, which significantly reduces the peak force the baffle must withstand. As the full momentum of the fluid is not transferred to the baffle as an impulsive load, the durability of baffle increases along with the prevention of sudden pressure spikes during violent sloshing.

The horizontal member of the baffle contributes to the suppression of the elevation of the wave, therefore, mitigating the impact in the direction of the vertical walls of the tank assisting in reduction of peak pressure and force.

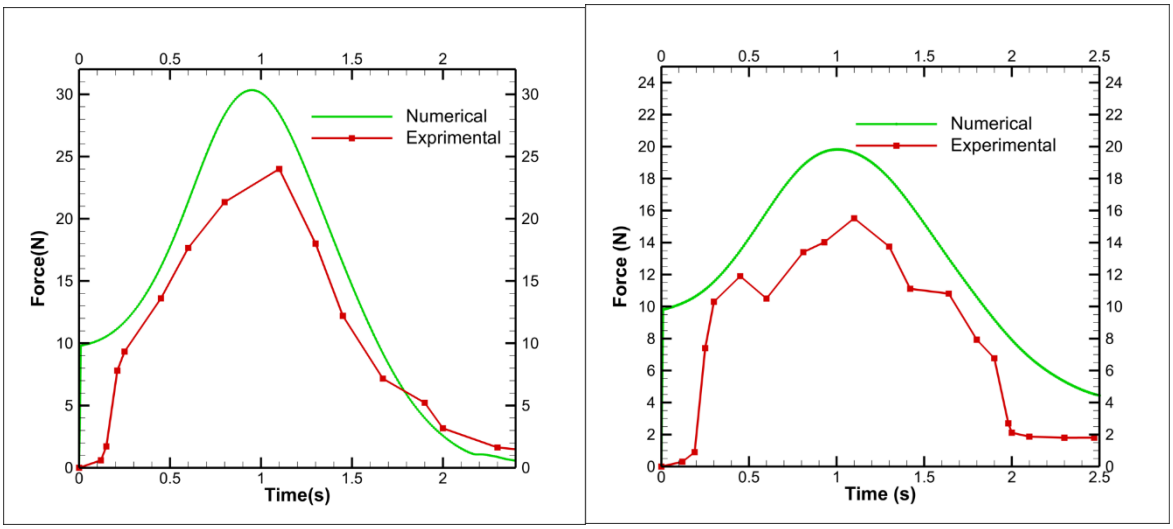
3.6 Comparison between experimental and numerical results

In present study, the numerical investigation was validated with an experimental study for the tank of rectangular cross section. The experimental setup was provided with a sinusoidal

motion which is same as the motion in the numerical studies with the help of crank wheel mechanism. Both qualitative and quantitative analysis was performed in the experiment.

3.6.1 Quantitative Comparison

The quantitative analysis was performed with a 10kg load cell along with amplifier HX711 and the reading was recorded on the computer connected to the load cell via Arduino Uno. In Figure 3.13, the quantitative comparison between experimental and the numerical study is presented.



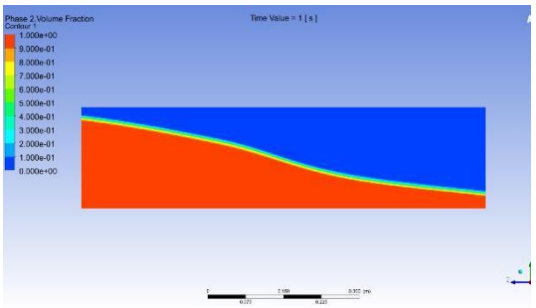
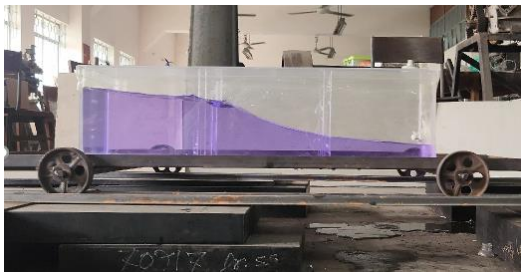
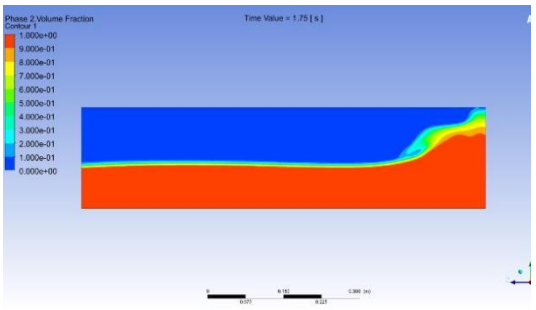
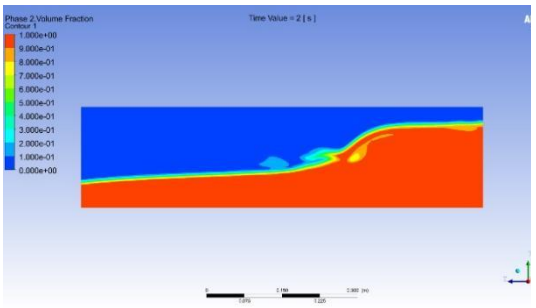
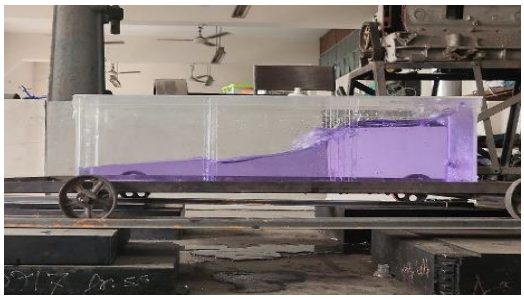
(a) **(b)**

Figure 3.13: The variation in experimental and numerical quantitative result for (a) Tank without baffle and (b) Tank with baffle

The time period for both the studies are same. There are some inconsistencies between both of the studies which are supposedly caused by the inertia of the experimental setup, noise and mechanical friction of the wheels on the rail of the setup which are not included in numerical analysis. As the force in our scale of model is very low overall compared to the actual scale, so, the effect of noise falls greatly upon the result in experimental analysis. Furthermore, the numerical analysis underestimates the viscous damping of the fluid and wall friction between fluid and tank wall.

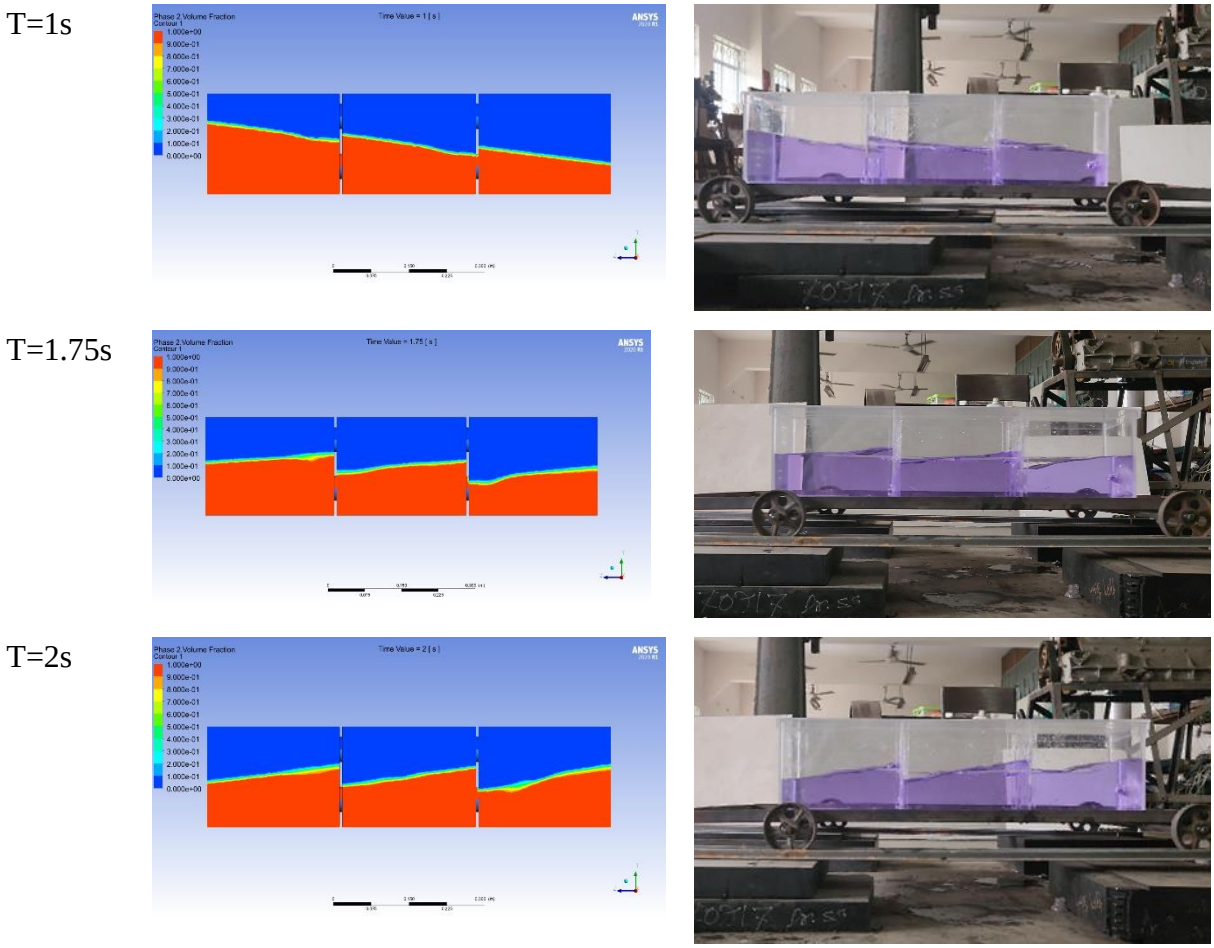
3.6.2 Qualitative Analysis

Still images of the experimental analysis has been captured for different flow time and was compared to the images from the numerical analysis. In the Figure 3.14, the images from both experimental and numerical analyses have been presented for baffle and unbaffled rectangular tank. The free surface elevation is almost same for both, although there are some factors like splashing of water droplet from noise has been detected in the experimental result.

Time	Numerical	Experimental
T=1s		
T=1.75s		
T=2s		

(a)

Time	Numerical	Experimental



(b)

Figure 3.14: The variation between numerical and experimental free surface elevation of rectangular tank (a) without baffle, (b) with baffle

3.7 Uncertainty analysis

Uncertainty analysis expresses how the errors in primary measurement propagate into the final result. In the present study, the load cell was calibrated with some known weights where there it uses a voltage value generated from the strain gauge deformation within the load cell to find the force applied to the load cell module. So, uncertainty exists in the voltage values along with the calibration parameter.

Table 3.1: 10 kg Load Cell Default Specifications

Specification Parameter	Absolute Value (for 10 kg / 98.1 N)
Rated Output (Sensitivity)	20 mV at full scale (with 10V excitation)
Combined Error (Linearity + Hysteresis) a_{Linear}	±0.002 kg (±0.0196N)
Repeatability, $a_{\mathrm{Re}\,peat}$	±0.001 kg (±0.0098N)
Creep (30 minutes), a_{creep}	±0.002 kg (±0.0196N)
Temperature Effect on Zero	±0.003 kg per 10ºC change
Temperature Effect on Span	±0.002% of whatever force is currently applied

For uniform distribution, the following equations are used to find the uncertainty

Linear uncertainty,

$$u_{Linear} = \frac{a_{Linearity}}{\sqrt{3}} \quad \textbf{3.1}$$

Repeatabe Uncertainty,

$$u_{\mathrm{Re}\,peat} = \frac{a_{\mathrm{Re}\,peat}}{\sqrt{3}} \quad \textbf{3.2}$$

Combined Uncertainty,

$$u_c = \sqrt{(u_{Linear})^2 + (u_{\mathrm{Re}\,peat})^2} \quad \textbf{3.3}$$

Drift Limit,

$$a_{dritf} = \frac{\boldsymbol{Force \times Temperature\ Effect\ on\ Zero}}{\boldsymbol{Temperature\ change\ for\ that\ force}} \quad \textbf{3.4}$$

Uncertainty in static baseline,

$$u_{V_0} = \frac{a_{drift}}{\sqrt{3}} \quad \textbf{3.5}$$

Uncertainty in the dynamic value,

$$u_v = \frac{a_{creep}}{\sqrt{3}} \quad \textbf{3.6}$$

Total uncertainty of the load cell,

$$u = \sqrt{(u_c)^2 + (u_{v_0})^2 + (u_v)^2} \quad \textbf{3.7}$$

Solving the equations above based on the specifications given in Table 3.1, We get the uncertainty of the load cell

$$u = \sqrt{(0.0126)^2 + (0.034)^2 + (0.0113)^2} = 0.0173N\,(approax)$$

Where, $u_c = 0.0126\,\text{N}$, $u_v = 0.0113$ N and $u_{v_0} = 0.034$ N.

Chapter 4

Broader Impacts

4.1 Health, Safety, Legal, Social and Cultural Issues

4.1.1 Health and Safety Hazards

The experimental part of this study involved a laboratory scale reciprocating motion setup, a motor driven wheel and connecting rod mechanism, electrical components and a water filled tanker model. The main safety hazards were related to moving mechanical parts, possible vibration of the platform, loose connections, water leakage and electrical contact. Since the setup produced repeated back and forth motion, accidental contact with the connecting rod, rotating wheel or moving platform could cause injury. Although water was used as the working fluid and no chemical hazard was involved, leakage near the motor, Arduino Uno, HX711 amplifier or other electrical connections could create an electrical safety risk.

4.1.2 Safety Measures and Personal Safety Measures Followed During Testing

Before each test, the tanker model was firmly fixed on the moving platform to prevent slipping or unwanted displacement. The connecting rod, rotating wheel, supporting frame and fasteners were checked to ensure that the setup was mechanically stable. Hands, tools and loose objects were kept away from the moving mechanism while the motor was operating. Electrical connections of the motor, load cell, HX711 amplifier and Arduino Uno were inspected before running the experiment. The tank was also checked for leakage so that water could not come into contact with electrical components. The experiment was performed at the selected controlled speed and the setup was continuously observed during operation. If abnormal vibration, leakage or mechanical looseness was noticed, the test was stopped immediately.

Standard practices for lab safety were observed throughout the experiment. The user stayed at a safe distance from the operating machine without interfering with it in any way. Proper footwear and lab attire were worn to minimize injury caused by any contact between the machine or spillage of water. The work area was also kept dry to avoid the possibility of slipping, falling, or electrical contact.

4.1.3 Social, Cultural and Practical Considerations

This study is socially relevant because sloshing in partially filled road tankers is directly related to transportation safety. When the liquid inside a tanker moves strongly during braking, acceleration or turning, it can reduce vehicle stability and increase the risk of accidents. A better understanding of tanker shape and baffle arrangement can help in designing safer tankers for liquid transport. This may benefit tanker drivers, transport companies, road users and people living near transport routes. In practical applications, reducing sloshing can also help prevent cargo loss, traffic disruption and environmental damage caused by tanker accidents.

No major cultural issue was directly related to this study. However, the practical use of the research findings is important especially for real tanker design and liquid transportation. In many transport systems, a sloshing control method should not only be effective but also simple, durable, affordable and easy to maintain. From this point of view, perforated baffles are a practical option because they can reduce liquid movement without using any complex control system. They can also be manufactured using common fabrication methods, making them suitable for local engineering and industrial applications. Therefore, this study supports a practical solution that can be adapted according to available materials, manufacturing facilities and operational needs.

4.2 Environmental Impact and Sustainability

The environmental impact of this study was relatively low because the experiment was carried out using a small laboratory scale model and water was used as the working fluid. No hazardous liquid, fuel or chemical substance was used during testing. Therefore, the experiment did not create any toxic waste or direct chemical pollution. The main environmental concerns were limited to electricity use for the motor and computer and the material used to fabricate the acrylic tanker model and baffles.

The use of numerical simulation also helped reduce environmental impact. By using ANSYS Fluent, different tanker shapes and baffle arrangements could be studied without fabricating every model physically. This reduced the need for extra materials, water use and repeated experimental trials. The experimental setup was also reusable, which made the work more resource efficient.

The findings of this study are environmentally relevant because sloshing can increase the risk of tanker instability and accidents during liquid transportation. If a tanker carrying fuel,

chemicals or other harmful liquids loses stability, it may lead to leakage, spillage and environmental contamination. A better understanding of tanker geometry and baffle arrangement can help improve tanker safety and reduce the chance of such incidents.

The use of perforated baffles is also a practical and sustainable mitigation method. Once installed inside the tanker, baffles do not require any additional power to operate. They work passively by interrupting liquid motion and reducing the intensity of sloshing. This makes them simpler, more durable and more energy efficient than active control systems.

From a lifecycle point of view, environmental impact mainly occurs during the manufacturing and disposal stages. In the experimental work, acrylic was used for model fabrication while in real tanker applications, metallic baffles would be more common. Metallic baffles can generally be reused or recycled at the end of service life, depending on the material and condition. Therefore, future full scale designs should consider material durability, maintenance requirements and recyclability.

This study is also connected to sustainable engineering practice. It supports safer transport design, helps reduce the possibility of cargo loss and may contribute to lower environmental risk in liquid transportation. Although the study does not directly measure carbon emission reduction, it provides useful design knowledge for safer and more sustainable tanker systems.

Chapter 5

Complex Engineering Problem Mapping

5.1 Introduction

The present thesis addresses a Complex Engineering Problem related to liquid sloshing in partially filled road tankers. Sloshing is a transient free surface flow phenomenon in which liquid motion is affected by tanker geometry, fill level, excitation condition, internal baffle arrangement, wall pressure, hydrodynamic force and practical transportation safety requirements. Therefore, the problem cannot be solved by a single empirical equation or a routine design procedure.

This chapter evaluates the present research in relation to the Washington Accord Complex Engineering Problem attributes P1–P7. The purpose of this mapping is to demonstrate that whether the thesis problem involves sufficient technical depth, analytical complexity, practical constraints, stakeholder relevance and system level interdependence or not.

5.2 CEP Mapping of the Present Study

Table 5.1: Assessment of the Present Research Based on Washington Accord Complex Engineering Problem Attributes

P code	Attribute	Applies	Justification and Evidence Location
P1	Depth of Knowledge	Yes	The study applied fluid mechanics, transient free surface flow, VOF based CFD modelling, mesh generation and load cell based experimental measurement. Evidence is presented in Chapter 2, especially Sections 2.2 and 2.3.
P3	Depth of Analysis	Yes	The problem required quantitative and qualitative analysis of pressure, force, free surface motion, baffle effect and numerical experimental comparison.

			Evidence is presented in Chapter 3, especially Sections 3.1–3.6.
P6	Stakeholder Diversity	Yes	The work is relevant to tanker drivers, road users, transport companies, tanker manufacturers, industries carrying liquid cargo and nearby communities. Evidence is discussed in Chapter 4 under health, safety, social, practical and environmental impacts.
P7	Interdependence	Yes	The study involved connected subsystems including tanker geometry, liquid motion, excitation mechanism, baffle arrangement, CFD model, experimental setup, sensor measurement and validation. Evidence is presented in Chapters 2 and 3.

5.3 Overall Justification of Complexity

Based on the above mapping the present thesis satisfies the requirements of a Complex Engineering Problem. The study meets the mandatory P1 attribute because it requires in depth engineering knowledge of fluid mechanics, numerical simulation, experimental measurement and sloshing mitigation. In addition, the study satisfies several other attributes such as P3, P6 and P7.

The investigation cannot be considered a routine engineering problem because the sloshing behavior of partially filled tankers changes with tanker geometry, excitation condition and internal baffle arrangement. The research therefore required a combined numerical and experimental approach to evaluate free surface behavior, force response and mitigation effectiveness.

In conclusion, the present study addresses a Complex Engineering Problem by investigating liquid sloshing in partially filled road tankers through CFD simulation, experimental validation and comparative design analysis. The findings support safer and more practical tanker design

by identifying how tanker geometry and perforated baffle arrangements influence sloshing reduction.

Chapter 6

Conclusion and Recommendations

6.1 Conclusion

The present study investigated sloshing behavior in partially filled capsule shaped, cylindrical and rectangular road tankers and evaluated the effectiveness of perforated baffles as a passive mitigation technique and also the objectives were successfully achieved through numerical simulation and experimental investigation. The main conclusions drawn from the results are given below:

1. The sloshing behavior of the selected tanker geometries was successfully investigated. The results showed that tanker shape strongly influenced the pressure and force response. Among the three geometries the capsule shaped tanker produced the highest sloshing response while the cylindrical tanker showed the lowest response under the same numerical condition.

2. The numerical and experimental flow contours of the rectangular tanker showed similar overall behavior. Both results showed larger free surface movement in the unbaffled condition and reduced liquid motion in the baffled condition. This confirms that the numerical model was able to represent the main physical behavior of sloshing.

3. The sloshing induced force was measured experimentally for the rectangular tanker using a load cell. The peak force was reduced from 24 N in the unbaffled case to 15.52 N in the baffled case. This gives a reduction of approximately 35.3% proving that the perforated baffle arrangement was effective in reducing sloshing induced force.

4. The effectiveness of perforated baffles was confirmed from both numerical and experimental observations. The baffles reduced liquid movement by dividing the flow region, interrupting wave propagation and dissipating part of the kinetic energy of the liquid. The perforations allowed controlled flow passage and helped reduce sudden pressure build up.

Overall, the study proved that both tanker geometry and perforated baffle arrangement have significant effects on sloshing behavior. The results support the use of suitable tank geometry

and internal perforated baffles for reducing sloshing-induced force and improving the stability of partially filled road tankers.

6.2 Recommendations and Future Work

Although the objectives of this study were achieved, some limitations can be addressed in future work. The experimental validation was performed only for the rectangular tanker. Therefore, capsule shaped and cylindrical tankers should also be tested experimentally for stronger validation.

Future studies should consider more realistic road conditions including sudden braking, cornering, combined longitudinal and lateral motion and road induced vibration. Different filling levels such as 25%, 50%, and 75%, should also be investigated because sloshing behavior changes with liquid depth.

Further work can also focus on optimizing the baffle design by varying the number of baffles, hole size, perforation ratio, baffle thickness and baffle position. In addition, multiple pressure sensors or load cells may be used to obtain more detailed force and pressure distribution on the tank walls.

Future simulations should include more detailed mesh and time step independence studies for all tanker geometries. Full scale tanker models and different working fluids such as fuel or industrial liquids, may also be considered to make the results more applicable to real transportation systems.

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